

SAI 3.2

System Analysis with AI

Target System: Laboratory Innovation Brain for Silicon Photonics

Human-readable, Agent-executable System Analysis & Implementation Specification

第一個垂直領域：Silicon Photonics

以矽光子 Ring Modulator / PN Junction 為第一個真實研究環境，系統核心維持 domain-agnostic；後續可擴充至 MZM、coupler、WDM、packaging、semiconductor devices 與其他工程科學領域。

SAI Document Version 3.2

2026-09-10

Implementation target: Silicon Photonics first | Core architecture remains domain-
extensible | System vision: 3-5+ years

文件控制與閱讀方式

項目	內容
文件名稱	SAI 3.2 - System Analysis with AI: Laboratory Innovation Brain for Silicon Photonics
版本	3.2
日期	2026-09-10
核心定位	Release Candidate Closure：Executable Epistemic Architecture 已定型，本版封閉規格 自洽、型別、運算子、migration 與 Agent entrypoint 缺口。
第一個 Domain	Silicon Photonics - Ring Modulator / PN Junction first vertical; adjacent domains are extension targets
主要讀者	Implementation Agent、教授、研究生、系統 開發者、AI for Science / photonics 研究者
設計時間尺度	M0a-M4 可逐步落地；M5+ 明確 DEFERRED； 完整 Laboratory Innovation Brain 願景 3-5+ 年演進。
核心原則	human-guidable, agent-executable, evidence-grounded, provenance-first, typed- tool, domain-modular, benchmark-driven

本文件屬於 SAI（System Analysis with AI）規格文件。SAI 是分析與實作規格的方法/文件名稱，不是系統產品名稱。被分析、被實作的目標系統暫稱 Laboratory Innovation Brain。文件採雙軌寫法：人類讀者先理解 Why / boundary / risk；Implementation Agent 則依 MUST / SHOULD / MAY、Requirement ID、Interface Contract、Acceptance Test 與 Repository Tree 實作。Silicon Photonics 是第一個落地 domain；核心不得因第一個 domain 而失去後續跨領域擴充能力。

v3.2 Revision Scope - Release Candidate Closure

v3.2 不改變 v3.2 的 Executable Epistemic Architecture；本版只做 release-candidate closure：修正 requirement/test traceability 自我一致性、補齊 EvidenceAuthority/TransitionPolicy 邊界行為、定義 BenchmarkPolicy、OutcomeSpace、ValidationReport 與 disagreement metric

contract、強化 high-stakes Critic inverted retrieval、補 migration/repository/spec-lint 缺口，並清除版本與懸空型別殘留。

Normative rule: SAI = System Analysis with AI document/specification. Target system = Laboratory Innovation Brain. Core MUST remain domain-agnostic; Silicon Photonics MUST be implemented as the first DomainPack. Agent MUST NOT silently alter architecture, schemas, evidence semantics, simulator authority, security boundaries, or requirement IDs without an Architecture Decision Record (ADR).

0.1 Terminology Boundary - SAI 與目標系統不可混用

本文件從 v2.0 起強制區分「規格文件」與「被實作系統」。這不是命名美學，而是為了讓 Agent 不會把文件版本、系統版本、DomainPack 版本與資料 Schema 版本混在一起。

名詞	正式定義	在程式碼/文件中的規則
SAI	System Analysis with AI；本文件/分析規格本身	只用於文件版本，例如 SAI 3.2；不得作為核心 Python package、DB schema 或 Agent 名稱。
Laboratory Innovation Brain	被分析、被實作的長期實驗室科研系統	可簡稱 Laboratory Brain；系統產品名稱未來可另定。
Silicon Photonics DomainPack	第一個實作領域	所有 Cj、Rs、ring、CHARGE 等知識必須位於 domain/plugin 或 simulator adapter，不得污染 domain-agnostic core。
System Version	實際程式/部署版本	與 SAI 文件版本獨立，例如 system v0.1.0 可由 SAI 3.2 規格實作。
Schema Version	Artifact/Evidence/Hypothesis 等資料契約版本	必須可 migration；不可用文件版本取代 schema version。

0.2 Dual-Audience Contract - 人類看 Why，Agent 看 What/How/Test

SAI 3.2 的每一項關鍵規格都應同時回答四件事：Why（人類理解目的與取捨）、Requirement（Agent 必須做到什麼）、Implementation Boundary（哪些可自行決定、哪些不可改）、Acceptance Test（如何證明完成）。沒有可驗收條件的描述只能視為 design note，不視為完成規格。

規範詞	Agent 解讀	人類如何使用
MUST / MUST NOT	硬性契約；不符合即不接受實作	用於 epistemic integrity、security、provenance、

規範詞	Agent 解讀	人類如何使用
		domain boundary、typed tools 等不可破壞規則。
SHOULD / SHOULD NOT	預設遵守；若偏離必須建立 ADR 說明原因	允許因環境、solver、基礎設施限制調整。
MAY	可選實作；不能阻塞核心 vertical slice	通常屬於優化、UI、scale-out、future domain。
NOTE	解釋與背景，不直接形成驗收門檻	讓人類知道為什麼這樣設計。

0.3 Agent Change-Control Rule

Agent MAY choose implementation details inside an approved boundary.

Agent MUST NOT silently change scientific semantics.

If a requirement must change:

1. create docs/adr/ADR-XXXX.md
2. state problem, alternatives, chosen change, migration, risks
3. mark affected Requirement IDs
4. wait for human approval when the change touches P0 contracts

P0 contracts 包含：Inference is not Evidence、artifact immutability、condition provenance、fidelity authority、simulator validity、external-data privacy boundary、core/domain dependency direction、以及 measurement/simulation 的來源分離。

0.4 Verification-in-the-Loop Core Baseline

核心原則：創新性大腦不是「自動多跑模擬」，而是知道什麼問題只需查已知證據、什麼問題可用解析計算或歷史案例回答、什麼問題值得啟動 surrogate / simulator、以及什麼問題必須留給真實量測或 human gate。Simulation 是 Silicon Photonics 的重要 backend，但不是 core ontology。

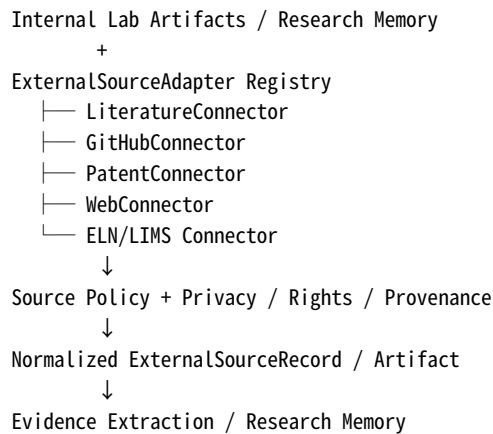
Human NL / Research Goal

- Problem Structuring
- Evidence Source Selection
- Independent Specialist Reasoning
- Competing Hypotheses
- Adversarial Critique
- Verification Planner
- Least-Cost Sufficient Evidence-Producing Action
- Observation / Evidence
- Belief Revision
- Novelty Audit
- Innovation Candidate + Verification Level

0.5 v3.2 External Source Connectivity Contract - Dataset / GitHub / Web 都經可替換 Adapter

Laboratory Innovation Brain 的創新材料來自內部自架 Knowledge Base 與外部 sources 的聯邦式組合。Core cognition 不得直接呼叫 GitHub、Crossref、Web 或特定 ELN SDK；它只依賴

ExternalSourceAdapter / SourceRouter contract。這使 GitHub 只是其中一個 provider，而不是系統硬依賴。



0.6 v3.2 Specification Closure Contract - 規格本身必須先通過自己的 CI

v3.2 的原則是「規格先可執行，再開始施工」。任何 normative schema、operator、Requirement ID、Test ID、migration、repository contract 與 Appendix 必須彼此一致；T-SPEC-001/002 若失敗，Agent MUST block 當前 slice、建立 spec issue/ADR，而不得自行猜測或選擇較像程式碼的段落。

Canonical v3.2 scientific state:

Artifact / SourceWork

- > Claim or Observation
- > Attestation
- > Provenanced RelationJudgment
- > TransitionPolicy evaluation
- > BeliefRevisionEvent (append-only)
- > EpistemicStateProjection (rebuildable)

Canonical execution:

Actor/ACL + EvidenceBundle + Capability + CostVector + Job + trace_id

Canonical cognition:

6 Core Roles + N Domain Specialists

Stake-adaptive debate + Critic inverted retrieval

Canonical verification:

Intent-aware SourcePolicy + AuthorityPolicy + deterministic plan selection

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1. Executive Summary - Laboratory Innovation Brain 是什麼

Laboratory Innovation Brain（本 SAI 文件的分析標的）是一套長期部署於研究實驗室的人工科研大腦。它的目標不是回答問題，而是把「實驗室已知什麼、為什麼知道、在哪些條件下成立、曾經被什麼證據支持或推翻、下一個最值得做的實驗是什麼」轉成可追蹤、可驗證、可演進的科研狀態。

核心定義

Laboratory Innovation Brain = Persistent Scientific Memory + Evidence-grounded Reasoning + Competing Hypotheses + Active Verification Planning + Deterministic Tools/Simulators + Belief Revision + Long-term Learning。

SAI 3.2 = 用來描述、約束、分解並驗收上述系統的 System Analysis with AI 規格文件。

系統第一個垂直領域選擇 Silicon Photonics，原因不是把 Laboratory Innovation Brain 寫死在光子學，而是矽光子具備非常好的「數位實驗環境」：CHARGE、MODE、FDTD、INTERCONNECT 等 simulator 能在 fabrication 之前提供大量 objective feedback，適合建立 AI hypothesis → simulation → observation → revision 的閉環。

圖 1. Laboratory Innovation Brain 總體視圖：人類 PI、科研記憶、認知功能、工具/模擬器、外部驗證與長期學習形成閉環。

1.1 Laboratory Innovation Brain 與一般 RAG / AI Scientist / Optimizer 的差異

系統類型	主要問題	記憶	外部驗證	Laboratory Innovation Brain 的差異
一般 RAG	「資料裡寫了什麼？」	文件 chunk / embedding	通常沒有	Laboratory Innovation Brain 將文獻轉成 condition-aware evidence，並連到 hypothesis / experiment。
聊天型 Multi-Agent	「多個角色怎麼討論？」	conversation history	常靠 LLM judge	Laboratory Innovation Brain 把 Agent 視為 cognitive function，

系統類型	主要問題	記憶	外部驗證	Laboratory Innovation Brain 的差異
				scientific truth 不由投票決定。
Autonomous Optimizer	「怎麼讓 FoM 更高？」	best design / journal	simulation score	Laboratory Innovation Brain 還問 why，維護 competing mechanisms，選實驗去區分它們。
AI Scientist workflow	「能不能自動走完整研究流程？」	task state / paper artifacts	多為 code/ML experiments	Laboratory Innovation Brain 以實驗室長期記憶與 scientific belief state 為核心，目標是跨年份持續成長。
Laboratory Innovation Brain	「實驗室目前相信什麼？哪些未知？下一個最有資訊價值的驗證是什麼？」	Artifacts + Evidence + Epistemic State	simulator / measurement / prior art	長期累積研究經驗、失敗與 tacit knowledge，並讓 evidence / hypothesis / experiment 形成可追溯研究閉環。

1.2 核心 scientific loop

Evidence → Competing Hypotheses → Criticism → Verification Planning
 → Evidence-Producing Action → Observation → Support / Contradict / Inconclusive
 → Belief Revision → Hypothesis Evolution → Memory → Next Cycle

這個 loop 的關鍵不是「讓 LLM 自我反思」，而是把改變信念的權力交給可追溯的外部 evidence。LLM 可以提出、解釋、挑戰、規劃，但不能把自己的上一輪文字當作下一輪 scientific evidence。

2. Problem Definition - 為什麼實驗室需要 Innovation Brain

實驗室的真正資產遠超過已發表 paper。教授與學長姐的 tacit knowledge、失敗 simulation、錯誤 mesh、未發表 parameter sweep、量測校正經驗、PDK 限制、meeting 決策與未完成的 research gap，往往才是下一個研究能否快速推進的關鍵。然而這些資訊通常分散在個人腦中、PPT、聊天紀錄、不同硬碟與 solver project 裡。

- 知識流失：學生畢業後，很多「踩坑經驗」只留下結果，不留下因果與診斷順序。
- 證據碎片化：paper、simulation、measurement 與教授評論沒有共同 ID 與 provenance。
- 研究容易重複：新學生可能重跑學長姐已經失敗過的 parameter space。
- RAG 不等於研究能力：找得到 paper 不代表能形成可證偽 hypothesis，更不代表知道下一個實驗怎麼選。
- LLM 自信問題：模型常把「合理敘述」誤當「已驗證事實」，若沒有 epistemic type 與 admission gate，長期記憶會被污染。
- 工程與科學混淆：找到更好的 FoM 並不等於理解 mechanism，也不一定能跨 device transfer。

Laboratory Innovation Brain 的最高價值
不是「少寫幾行 Lumerical script」，而是讓實驗室在 3-5 年後仍能檢索、重現並利用過去幾千次研究決策與失敗，形成可傳承、可累積的科研能力。

2.1 長期目標與非目標

類別	目標系統要做	Laboratory Innovation Brain 不應假裝做到
Scientific memory	保存證據、條件、研究 episode、假說演化、專家 heuristic	把聊天全文永久塞進 vector DB 就叫 memory
Scientific reasoning	提出多個 competing hypotheses，明示 falsifier 與 confounder	用一個 LLM judge 分數宣告某假說為真
Verification planning	選擇最有區辨力、成本合理的下一步驗證	只做 greedy simulator/FoM optimization
Automation	透過 typed tools 控制模擬/資料分析	讓 LLM 任意 eval 任意 script 並可改 evaluator
Learning	先用 memory/case retrieval，再蒐集 trajectories 做週期性訓練	每跑一次 simulation 就 online gradient update
Human role	PI 設方向、限制、保密與高成本 gate	宣稱完全取代教授或 fabrication expert

3. Design Principles - 不可違反的系統原則

P1 - Evidence over Eloquence

任何 scientific claim 必須可追溯到 artifact、文獻 locator、simulation run 或 measurement；語言流暢不能提升證據等級。

P2 - Artifact Immutability

原始資料預設 append-only，以 content hash 綁定。解析結果可以重建，但 raw artifact 不應被 AI 偷改。

P3 - Inference is not Evidence

LLM 產生的 interpretation、hypothesis、speculation 永遠保留 epistemic type，不可自動晉升為 observed fact。

P4 - Condition-aware Science

矽光子不存在脫離 platform、geometry、bias、wavelength、temperature、doping 的孤立數值。Evidence 必須保存適用條件。

P5 - Agent is a Cognitive Function

多 Agent 代表功能分工，不代表每一角色必須是一顆不同 LLM。

P6 - Deterministic Tools, Reasoning Models

模擬器與 metric extractor 是可重現工具；LLM 負責提出問題與選擇動作，不直接擁有 scientific ground truth。

P7 - Evaluator First

先建立可信 evaluator / parser / provenance，再擴大 Agent swarm。

P8 - Negative Results Are Assets

失敗不刪除；保存 symptom、candidate cause、confirmed root cause、fix 與 prevention rule。

P9 - Private by Construction

未發表 idea、NDA PDK、internal measurement 預設不離開 local boundary；外部搜尋經 policy gate。

P10 - Scale by Evidence

是否抽離 vector/graph service 以 benchmark 決定，不用「十萬向量」「五百萬節點」這類硬門檻。

P11 - Human Governance for Irreversible Cost

fabrication、昂貴 solver budget、對外資料傳輸與 IP 相關步驟必須有明確核准 policy。

P12 - Core Domain-Agnostic, Domain Pack Specific

核心處理 evidence / hypothesis / episodes / tools；矽光子物理規則、metrics 與 Lumerical adapter 放在 domain pack。

3.1 v1.1 新增不可違反原則

P13 - Unknown Is a Valid Scientific State：抽取器無法確認的參數必須標記 UNKNOWN / NOT_REPORTED / REFERENCED_EXTERNALLY / CONFLICTING；禁止為了填滿 schema 而推測缺失值。

P14 - Fidelity Determines Epistemic Authority：Exploratory fidelity 可以排序與提出疑點，但不能單獨形成 confirmed contradiction；重大 hypothesis rejection 至少需要 standard/validation fidelity 或可重現的實體量測。

P15 - Simulator Is Evidence, Not Ground Truth：模擬結果必須帶 solver convergence、mesh sensitivity、model version、calibration/validation range 與 known limitations。Simulator/measurement 不一致時保存 conflict，不自動選邊站。

P16 - Low-Friction Knowledge Capture：Tacit knowledge 的結構化由 Miner 草擬，研究人員主要負責 approve/correct/reject。未經授權的聊天/錄音不得被被動監聽或自動納入。

P17 - Verification Backend Is Pluggable：Core 只理解 Evidence-Producing Action / capability / cost / authority；不得把 Lumerical、DFT、training run 或 wet-lab experiment 寫死為唯一驗證方式。

P18 - Least-Cost Sufficient Verification：系統目標不是最大化實驗次數，而是在能改變當前 scientific decision 的前提下，以最低合理成本取得足夠證據。

P19 - Evidence-Producing Action Before Simulator-by-Default：優先重用已存在 evidence、歷史 episode、解析/數值檢查與 surrogate；只有在不足以區分 hypotheses 時才升級至 simulator / physical experiment。

P20 - Unverified Is a Valid Output：如果目前沒有適當工具、權限或預算，系統可以輸出 UNVERIFIED INNOVATION CANDIDATE，明確列出缺少的驗證能力，而不是硬湊 simulation conclusion。

3.2 v3.0 新增不可違反原則

P21 - Event-Sourced Belief：EpistemicState 只是一個可重建 projection；任何 belief/status 變化都必須由 append-only BeliefRevisionEvent 表達。

P22 - Claim Identity before Corroboration：先判斷「是不是同一個 claim / 同一份 underlying work」，再計算有幾份獨立 evidence。preprint、正式版、review 轉述不得被默認成三份獨立支持。

P23 - Inference Provenance Everywhere：所有 LLM 產物與 LLM 形成的關係判斷，都必須保存 model/version、logical slot、prompt version、evidence bundle hash、retrieval policy 與時間。

P24 - Intent-Aware Retrieval：Source Router 不採固定來源排行榜；diagnosis、mechanism discovery、novelty audit、cross-domain innovation、replication 使用不同 SourcePolicy。

P25 - Cost and Waiting Are First-Class：wall-clock、human attention、money、license-seat time、calendar latency 與 irreversibility 都是 verification cost；長時工作必須可 suspend/resume。

P26 - Backend-Agnostic Metrics：run_* 綁定 simulator/measurement backend；extract_* 只依賴 typed numerical data + conditions，讓 simulation 與 measurement 使用同一定義比較。

P27 - Evidence Authority, Not Simulator Ladder：core 只知道 EvidenceAuthority 偏序/比較介面；mesh、solver convergence、calibration 等由 DomainPack/backend validity schema 擁有。

P28 - Default-Deny Sensitive Ingestion：未明確分類的新 internal artifact 以較嚴格 sensitivity 進入，經授權後才能降級；不得 fail-open。

4. Open-source Architecture Study - GitHub 取材與邊界

Laboratory Innovation Brain 不應 fork 一個現成 AI Scientist 後改 prompt。比較合理的是「組件式吸收」：從不同專案取已被驗證的 architecture pattern，再以 Laboratory Innovation Brain 的科研記憶與 hypothesis-testing 需求重新組合。下列專案是本版架構的主要參考。

參考專案	取材領域	建議採用	明確不照搬
flexcompute/ autophotonicdesign	Photonics autonomous execution	Explore → Design → DRC → Simulate → Analyze → Keep/Discard → Journal；有 PN modulator branch。	不把 FoM keep/discard 當 scientific discovery；不照搬單 Agent 工程最佳化。
Kaimen-Inc/Co-Scientist (Google AI Co-Scientist reimplementation)	Hypothesis search	Supervisor、Generation、Reflection、Ranking、Evolution、Proximity、Meta-review；pairwise/Elo 可用於 allocate attention。	Elo 不決定真理；要改成 simulator/evidence adjudication。
Future-House/paper-qa	Scientific evidence retrieval	scientific search、evidence gathering、citation grounding、reranking。	不是 Laboratory Memory；輸出後還需 evidence normalization / condition extraction。
lamm-mit/SciAgentsDiscovery	Graph-guided cross-domain reasoning	knowledge graph 與多角色 reasoning，可用於找非直接近鄰連結。	Phase 1 不必一開始建立大型 ontology 或專用 graph DB。
SakanaAI/AI-Scientist-v2	Agentic tree exploration	Progressive / best-first tree search、idea → experiment branching。	Photonics simulation 昂貴，需 budget-aware / multi-fidelity；授權條款需獨立注意。
zou-group/virtual-lab	Research-team protocol	模擬 PI / specialist / team meetings，適合參考 structured collaboration。	避免自由聊天與 groupthink；使用獨立立場、cross-examination。
SamuelSchmidgall/ AgentLaboratory	Workflow separation	Literature → Experiment → Report 分階段與 checkpoint。	不需要複製 end-to-end paper writing 為 Laboratory Innovation Brain 核心。
stanford-oval/storm	Perspective-guided questions	多視角提問、共享 conceptual space / mind map。	避免和 KG 重複；主要用於 gap discovery / research questioning。
Future-House/robin	Experiment-feedback scientific discovery	hypothesis / assay / candidate ranking + experimental data feedback。	生醫流程不可直接搬到 photonics；借 feedback-loop concept。
amair-lab/PriM	Principle-guided multi-agent discovery	literature → hypothesis → virtual experiment → optimizer/MCTS；重視物理/化學 principles。	不要把 property optimizer 直接當 hypothesis discriminator。
kotama7/ARI	Autonomous research infrastructure	BFTS、MCP plugin、claim-evidence gate、reproducibility / HPC pattern。	大型 paper automation pipeline

參考專案	取材領域	建議採用	明確不照搬
			非 Phase-1 必要。
leisymqaz/lumerical-mcp	Lumerical bridge	透過 lumapi 控制 FDTD / MODE / INTERCONNECT / DEVICE，persistent session。	Laboratory Innovation Brain 應自己包 typed scientific tools，不直接暴露 arbitrary eval_script；授權需獨立確認。
docling-project/docling	Scientific document ingestion	layout-aware PDF/Office 解析、tables/structure。	解析器是 adapter，不應綁死核心資料格式。
langchain-ai/langgraph	Stateful agent orchestration	explicit state graph、checkpoint、resilient workflow。	不要讓整份 raw artifact 進 agent state；state 只放 ID/summary。

4.1 Laboratory Innovation Brain 的組裝策略

Co-Scientist-style hypothesis search

- + PaperQA2-style evidence retrieval
- + SciAgents-style cross-domain connection
- + provenance / evidence-graph principles
- + AI-Scientist-v2-style budgeted branching
- + AutoPhotonicDesign-style photonic execution loop
- + SAI typed Lumerical tools
- + SAI Artifact/Evidence/Epistemic Memory
- = Laboratory Innovation Brain

真正需要自己設計的核心，不是 connector 數量，而是：scientific evidence 如何進入 hypothesis state、competing hypotheses 如何被外部實驗區分、失敗如何回寫、以及多年 research episodes 如何在下一次研究中被主動調用。

5. Laboratory Innovation Brain Overall Architecture - 系統總體架構

Laboratory Innovation Brain 採五個邏輯平面：Governance Plane、Scientific Memory Plane、Cognitive Plane、Execution Plane、Learning & Evaluation Plane。這些是邏輯分層，不等同五個 microservices；v1 可以在同一個 Python monorepo + PostgreSQL 中運行。

平面	責任	v1 實作重點
Governance	Research contract、project scope、privacy mode、budget、human approvals	本地 policy engine + session/episode state
Scientific Memory	Artifact/Evidence/Epistemic state、episode lineage、provenance	PostgreSQL + object abstraction + numerical arrays
Cognitive	Evidence、Hypothesis、specialist physics、critic、experiment design	同一/多模型 routing，structured outputs
Execution	Search、Code、Lumerical、metric extraction、plotting	typed tool adapters + immutable run manifests
Learning & Evaluation	benchmark、case retrieval、trajectory datasets、model improvement	先 benchmark，再訓練；不做即時 online learning

圖 2. Laboratory Innovation Brain 五平面互動：Memory 是中心，Agent 與工具都以 evidence/provenance 回寫。

5.1 核心狀態物件

- ResearchProject：長期研究主題、權限、DomainPack、budget 與資料邊界。
- ResearchEpisode：一次有明確 goal、state transitions、jobs、decision 與 outcome 的研究演繹。
- Artifact：不可變原始/衍生研究物件；content hash 是跨系統 provenance anchor。
- Evidence / Attestation：某個來源對 Claim/Observation 的具名、帶條件見證；不等於 claim 本身。
- Hypothesis：可證偽的 mechanistic proposition，帶 predictions / falsifier / assumptions；所有生成來源含 InferenceProvenance。
- VerificationPlan：為改變 scientific decision 而設計的 action portfolio，使用 Capability + CostVector + EvidenceAuthority。

- Run / Job：Run 是已執行 activity record；Job 是可排程、可 retry、可 suspend/resume 的執行單位。
- Observation：artifact/run 產生的客觀數值/事件；與 Claim、Interpretation 分離。
- Interpretation / Relation Judgment：對 observation/claim 的推論；必須標記 inferred 並帶 InferenceProvenance。
- Heuristic：人類或系統從多 episodes 抽出的診斷/設計經驗；需來源、scope、validated_on 與 approval。
- Claim：可被支持/反駁/轉引的科學命題 identity；與來源文件、Evidence 分離。
- BeliefRevisionEvent：append-only 科研狀態轉移事件；記錄 from/to state、觸發 evidence、policy/model/actor。
- EpistemicState：由 events + admitted relations 投影得到的「現在相信什麼」；不可作為不可追溯的 mutable truth row。
- Actor / Principal：人類、service account、Agent role 的可識別主體；ACL、approval、egress、audit 都依賴它。
- Capability：任何 evidence-producing backend 的機器可讀能力描述，包含 inputs/outputs/authority/cost/license/availability。

6. Knowledge & Memory Architecture - 資料庫、知識與記憶

本章採用外部對抗式審查最重要的修正：捨棄「Failure DB、Hypothesis DB、Evidence DB、Design DB 各自分開」的做法，收斂為三層 epistemic architecture。這可以避免多 store 同步與 ontology 在早期就僵化。

圖 3. Laboratory Brain Knowledge Core：聯邦式來源先進 Immutable Artifact Layer，再正規化 Evidence，最後形成可演進 Epistemic State。

6.1 Layer 1 - Immutable Artifact Layer

Artifact 是「曾經存在或發生過的原始資產」。它不需要被 LLM 完全理解才有保存價值。每個 artifact 由 content hash、source、sensitivity、timestamp、owner/project、format、parser version 等 metadata 描述。

Artifact 類型	例子	保存原則
Documents	PDF、PPTX、DOCX、meeting notes、thesis、patent	原始檔 + hash；解析結果另存 derived artifact
Simulation projects	.ldev / .lms / .fsp / .icp / .lsf / Python scripts	project file、solver version、Git commit、input manifest
Numerical arrays	C-V、S-parameter、spectrum、waveform、carrier grids	Zarr / Parquet / HDF5；不轉成自然語言當 primary data
Images	eye diagram、SEM、layout preview、field plot	原圖 + caption/context；若有 raw waveform 必須綁定
Measurements	CSV、Touchstone、instrument export	raw data + calibration state + instrument metadata
Web / external	合法取得 paper、metadata、HTML snapshot	依 license/rights policy 保存，不宣稱固定 fair-use 規則

6.2 Layer 2 - Scientific Assertion & Evidence Layer

v3.0 將「命題」與「證據」分離。Claim/Observation 表達「什麼被主張或觀察到」；Attestation/Evidence 表達「哪個來源、在什麼條件與 locator 下支持/報告它」。這能避免同一研究的 preprint、正式版、review 轉述被誤算為三份獨立 corroboration。

Claim / Observation = proposition identity
Attestation / Evidence = Source + Locator + Conditions + Method + Provenance
Relation Judgment = SUPPORTS / CONTRADICTS + InferenceProvenance
NOT: Evidence = free-form LLM summary or duplicated paper mention

- Condition-first：platform、device、geometry、doping、bias、wavelength、temperature、method。
- Locator-first：Paper page / figure / table、run_id、artifact hash、array slice。
- Epistemic-type：observed、measured、simulated、reported、human_asserted、inferred。
- Conflict-aware：SUPPORTS / CONTRADICTS 不存成 Evidence 的裸陣列；統一存為 Relation Judgment，且判斷者/模型/證據 bundle 必須可追溯。
- No self-citation loop：LLM 產生的 interpretation 可進 Epistemic State，但不可直接當 Evidence。

6.3 Layer 3 - Event-Sourced Epistemic State Layer

EpistemicState 是 Laboratory Innovation Brain 的 current scientific projection，不是唯一真實來源。所有狀態改變先寫入 BeliefRevisionEvent，再由 projector 重建 ACTIVE/SUPPORTED/CHALLENGED/REJECTED、open conflicts、research priorities 與 supersession。任何 extractor bug、retraction、錯誤 run 被隔離後，系統 MUST 能從事件與有效 evidence 重新投影 belief。

State	意義	例子
ACTIVE	仍在競爭、值得測試	H1: bandwidth improvement dominated by Cj reduction
SUPPORTED	目前 evidence 傾向支持，但非永恆真理	多組 bias sweep + matched-condition paper 支持
CONTRADICTED	關鍵 prediction 被高品質 evidence 反駁	控制 Cj 後 BW 仍大幅提升
INCONCLUSIVE	資料不足或 confounder 未排除	Rs 與 optical overlap 同時變動
SUPERSEDED	被更精確的 mechanism/model 取代	H1 → H1.2 增加 Rs / overlap interaction
DEPRECATED	因 calibration / model / process 過時	舊 material model 的 simulation conclusions

6.4 四類核心 Memory Taxonomy

核心記憶	包含	重要說明
Declarative Domain Memory	物理定律、材料常數、PDK 元件、已知 mechanism、標準術語	相對穩定；仍要有來源與版本
Episodic Research Memory	goal → hypothesis → plan → run → observation → decision	Failure / best design 都是 episode 中的 outcome/analysis，不需獨立 DB
Procedural Memory	typed tool workflow、SOP、parser、metric extractor、calibration recipe	最好是可執行程式 + environment manifest，不只文字 SOP
Epistemic Belief Memory	hypotheses、heuristics、conflicts、	Laboratory Innovation Brain

核心記憶	包含	重要說明
	open questions、support/contradict relations	的動態科研狀態；每次更新有 evidence lineage

6.5 Federated Data Sources - 自架與網路混合

Laboratory Innovation Brain 的資料來源應是 federated，而不是把所有東西先下載到單一 vector DB。Private Lab Knowledge、External Scientific Knowledge、Live Tool Output 各有自己的 source adapter 與 policy。

來源	典型內容	預設策略
Private Lab	thesis、meeting、PDK、unpublished result、simulation、measurement	local-first；敏感資料不外流
External Metadata	DOI、title、abstract、citation graph、patent metadata	可永久保留 metadata
External Full Text	paper / standard / docs	依授權與 access rights 決定 cache/retention
Broad Web	technical pages、standards pages、public web discovery	snapshot + timestamp + low/medium trust；主要用於探索，不自動升級為 scientific evidence
Live Simulator	CHARGE / MODE / INTERCONNECT outputs	立即生成 immutable Run + Observation
Human Expert	professor comment、diagnostic rule	標明 author、context、validation history；不是 peer-reviewed fact
GitHub / Remote Code Hosting	public/approved-private repos、README、code、commit、release、issue metadata	經 GitHubConnector / ExternalSourceAdapter；保存 repo + ref/commit + snapshot provenance；與 peer-reviewed evidence 分級

6.6 Storage Stack - v1 先求一致性，不追求微服務炫技

資料責任	Phase-1 建議	未來擴充條件
Metadata / relations / hypotheses / evidence	PostgreSQL + JSONB	保持 primary source of truth
Embeddings	pgvector in PostgreSQL	若實際 benchmark 顯示 latency/concurrency/memory 不足，再抽離 Qdrant 等 vector service

資料責任	Phase-1 建議	未來擴充條件
Graph relations	PostgreSQL edge tables + Recursive CTE	當多跳 traversal / graph algorithms 成實際瓶頸，再引入 Neo4j/Memgraph 等
Raw artifact	S3-compatible abstraction 或 NAS/local object layer	避免依賴單一 product；可替換 self-hosted / cloud backend
Large numerical arrays	Zarr / Parquet / HDF5 on artifact store	DuckDB / Python 做 in-place analysis；不逐點塞 SQL
Code/version	Git + content hashes + environment manifests	必要時加入 DVC/AiiDA-like provenance pattern

重要修正

Laboratory Innovation Brain 不把 MinIO、Neo4j、Qdrant 寫死，也不設定「超過某固定節點數就必須換 DB」的假精確門檻。所有 scale-out 由實際 benchmark 驅動。

6.7 Scientific RAG - Structure-aware + Evidence-aware + Condition-aware

Document / Artifact

- layout-aware parse
- semantic units (section / table / figure / equation / caption)
- evidence extraction
- condition normalization
- hybrid indexing (dense + sparse + structured filters)
- retrieval
- rerank
- context assembly
- Evidence Objects

固定 token chunk 只能作 fallback。對 scientific documents，更重要的是 evidence boundary：claim 與 supporting table/figure/caption/condition 不應被切散。

6.8 Condition-aware Retrieval

同一個「ring modulator bandwidth」可能在 O-band、C-band、不同 radius、不同 doping、不同 driver load、不同 temperature 下得到完全不同數值。因此 retrieval 應先做適用條件匹配，再做 semantic ranking。

Condition dimension	例子	用途
Platform	SOI 220 nm / SiN / foundry generation	避免跨平台硬比
Geometry	radius、waveguide width、slab、gap	判斷 device comparability
Electrical	doping、junction type、bias、load	比較 Cj / Rs / bandwidth
Optical	wavelength、polarization、Q / coupling regime	比較 resonance / loss / ER
Thermal	temperature、heater state	處理 drift / locking
Method	measurement / CHARGE / MODE / analytical	區分 evidence directness

6.9 Epistemic Provenance - 防止 LLM 自我引用污染

Epistemic type	可否直接當 Evidence	例子
MEASURED	可以	實驗室量測 S21 / spectrum
SIMULATED	可以，但標明 simulator/model	CHARGE Cj(V)、MODE neff
REPORTED	可以，需 source locator	paper Table II 數值
HUMAN_ASSERTED	可作 heuristic / prior，不等同 verified fact	教授：先查 contact continuity
INFERRED	不可以直接晉升	LLM：Cj 可能是主因
HYPOTHESIS	不可以	可證偽 mechanism proposition
SPECULATION	不可以	尚未形成 testable prediction 的 idea

6.10 Failure Analysis 是 first-class object，不是獨立資料庫

Failure 應保存在 ResearchEpisode 中，但其 Root Cause Analysis 需要結構化，因為未來 case-based reasoning 會依 symptom / context / cause / fix 找相似案例。

```
FailureAnalysis {
  failure_analysis_id, episode_id,
  symptom, expected_behavior, observed_behavior,
  candidate_causes[], confirmed_root_cause?,
  evidence_refs[], fix?, prevention_rule?,
  failure_class, resolution_status
}
```

6.11 Expert Heuristic / Tacit Knowledge

教授與學長姐的「直覺」不應被迫轉成真/假 claim，而應保存為 diagnostic heuristic：Trigger → Suggested Checks → Rationale → Source → Validated Cases。這類 memory 在異常診斷與實驗規劃時比一般文獻 RAG 更貼近實驗室價值。

6.12 ELN / LIMS 整合原則

Laboratory Innovation Brain 不應重新發明所有電子實驗筆記/樣品管理功能。正確方向是提供 adapters / webhook / file watchers，從現有 Git、雲端硬碟、ELN、solver output directories 非侵入式擷取資料。研究員只需在少數必要節點補充 intent / decision / root cause，而不是每天人工填數十欄。

6.13 Progressive Evidence Enrichment & Extraction Abstention

外部 scientific evidence 不應在 ingestion 第一輪被迫填滿所有條件。論文中的幾何、doping、bias、temperature、driver/load、fabrication process 常分散於正文、caption、Supplementary Information 或被引用文獻。SAI 3.2 規格要求目標系統採漸進式補齊：先保存可明確定位的 evidence，只有當該 evidence 真正進入 hypothesis discrimination 或定量比較時，才觸發 targeted enrichment。

- Stage A - Shallow ingest: metadata + section/figure/table anchors + 明確報告的核心數值。
- Stage B - Evidence extraction: 形成 claim/observation + conditions + locator。
- Stage C - Targeted enrichment: 只針對目前研究問題缺少的條件回查正文/SI/引用來源。
- Stage D - Human review: 若關鍵欄位仍 conflicting 或不可確定，升級給研究者，而不是由 LLM 猜值。

每個可條件化欄位都使用 field-level epistemic status：EXPLICIT（來源明示）、DERIVED（可由來源中的明確數值/公式計算）、INFERRED（模型推論，不能當原始 evidence）、UNKNOWN/NOT_REPORTED（來源未提供）、REFERENCED_EXTERNALLY（原文把條件委託給另一來源）、CONFLICTING（同一來源或多來源互相衝突）。這些狀態比單一 extraction confidence 更重要。

Abstention contract: Evidence extractor 可以輸出「不知道」，且 UNKNOWN 不降低 artifact 保存價值；只有當缺失條件會改變 scientific decision 時，才進一步補齊。

6.14 GraphRepository Abstraction - PostgreSQL-first，但不綁死查圖後端

v1 的 relations 仍以 PostgreSQL 作為 Single Source of Truth；但所有多跳關係查詢必須經 GraphRepository 介面，而不是讓各 Agent / service 直接散落 Raw Recursive CTE。這不是為了 Day 1 換圖資料庫，而是避免未來 ontology 與 traversal backend 綁死在 business logic。

GraphRepository minimal interface:

- neighbors(entity_id, relation_types?, depth=1)
- find_paths(source, target, constraints?)
- trace_provenance(entity_id)
- find_support_chain(hypothesis_id)
- find_contradictions(hypothesis_id)
- find_related_failures(symptom/context)
- upsert_relation(relation)

Backend v1: PostgreSQLGraphRepository

Future optional backends: embedded/property-graph engine or dedicated graph service, only after workload benchmark proves need.

Migration policy 不使用「超過 N 個 nodes/edges 就換 DB」的固定門檻。觸發條件應來自真實 workload，例如 P95 多跳查詢延遲、DB CPU/IO 佔比、並發量、特定 graph algorithm 需求或維運成本。Repository abstraction 只能降低替換成本，不能宣稱 graph backend migration 是零成本。

6.15 Heuristic Miner - Tacit Knowledge 的低摩擦沉澱

教授與資深成員通常只在 meeting、週報、批註或已授權群組中留下短句經驗。Laboratory Innovation Brain 不要求他們填完整 Heuristic Form；Heuristic Miner 從「已授權輸入來源」自動草擬 Candidate Heuristic，再交由人類核准。

Authorized source → Candidate Heuristic → Provenance check → Human Approve / Correct / Reject → Active Heuristic

Candidate Heuristic 只可作為 suggestion，不可在未核准前提升為 lab rule。若其來源只是一位研究員的單次看法，status 應維持 PROVISIONAL；若被多次 episode 驗證，可提升為 VALIDATED_HEURISTIC。

Privacy rule: 系統只能處理明確納入治理範圍的 meeting transcript、minutes、weekly report、ELN/comment export 或 explicit connected channel；不得把「監聽所有聊天」當預設功能。

6.16 External Source Adapter & GitHub Connector - 創新資料來源的正式接口

自架資料庫是 Laboratory Innovation Brain 的長期研究記憶與 evidence store；GitHub、期刊/論文、專利、Web、ELN/LIMS 則是可插拔的外部資料來源。任何 cognition role 若需要外部資料，必須先向 Evidence Source Router 提交「需要什麼 evidence」，由 Router 經 policy engine 決定可用來源，再由對應 ExternalSourceAdapter 執行。Agent 不可直接把 provider SDK 寫進 hypothesis / critic / novelty code。

ExternalSourceAdapter family:

- LiteratureConnector: paper metadata/full text where authorized, DOI/citation locators
- GitHubConnector: public/approved-private repositories, code, README, issues/releases/commits as technical/prior-art sources
- PatentConnector: patent metadata, claims/assignee/family locators where available
- WebConnector: controlled broad discovery / technical pages
- ELN/LIMSConnector: approved laboratory systems
- LocalGitProvenance: local repository commit/hash binding; NOT the same function as GitHub retrieval

GitHub 的主要用途有三種：(1) Architecture/implementation discovery：找可重用工具與系統架構；(2) Engineering evidence：查公開程式、README、issue、release、技術範例；(3) Prior-art/novelty supporting source：確認公開 implementation 是否已存在。GitHub evidence 的 epistemic type 必須與 peer-reviewed paper / measurement 分開，不能因為 repo 有 stars 或 README 宣稱成功就升級為 scientific truth。

GitHub access modes:

PUBLIC_DISCOVERY - 無私有 context，搜尋/讀取公開 repository。

APPROVED_PRIVATE_REPO - 僅在已設定 credentials、allowlist、project scope 與 sensitivity policy 時讀取授權 private repo。

SNAPSHOT_REPRODUCIBLE - 對真正被引用的 repo/file 固定 commit SHA / tag / release ref，產生 snapshot hash；後續 reasoning 以 snapshot 為 provenance anchor。

每個 external record 至少保存 provider、source_type、canonical_locator、retrieved_at、visibility、rights/license metadata（若可得）、version/ref、content hash/snapshot hash、sensitivity、trust class 與 raw artifact pointer。缺少 license/rights metadata 時不得由 Agent 自行宣告可複製程式碼；可閱讀/分析不等於可直接 copy。

SRC-001: Core cognition MUST access external knowledge through ExternalSourceAdapter/SourceRouter, not provider-specific SDKs.

SRC-002: Every admitted external source MUST preserve canonical locator + retrieval time + provider + source type + provenance pointer.

GH-001: GitHubConnector MUST support public repository discovery and commit/ref-pinned file retrieval through a provider-independent contract.

GH-002: Private repository access MUST require explicit configured authentication, allowlist/project scope and security policy; restricted context MUST NOT be used to construct uncontrolled external queries.

GH-003: GitHub-derived claims MUST remain technical/prior-art evidence unless independently supported by scientific evidence; commit/ref and repository identity MUST be preserved.

GH-004: Referenced GitHub content SHOULD be snapshotted/cache-pinned when it materially supports a decision, benchmark, architecture choice or reproducibility claim.

6.17 Claim Identity、Attestation 與 Evidence Independence

ClaimResolver 必須先做 source-work resolution（例如 arXiv ↔ DOI 正式版、引用/轉述）與 proposition normalization，再建立 Attestation。Corroboration 只計算符合 TransitionPolicy 指定 independence_basis 的獨立見證。DEPENDENCE_UNKNOWN 對 independent-attestation

count 的貢獻固定為 0，保留為 supporting context，但不得滿足

min_independent_attestations；若因此無法達門檻，回傳 NEED_MORE_EVIDENCE 或 NEED_HUMAN_REVIEW，不得猜測為獨立。

```
Claim { claim_id, normalized_proposition, scope{}, domain, created_at }
Attestation { attestation_id, claim_id, source_artifact_id/source_work_id, locator, conditions{}, method{},
epistemic_type, extraction_provenance }
EvidenceIndependence { a_id, b_id, relation: INDEPENDENT|DERIVED_FROM|SAME_WORK|CITES|UNKNOWN, rationale_ref }
```

6.18 Belief Replay / Contamination Rollback

Extractor、parser、paper status 或 simulation validity 一旦被判定有問題，系統先 quarantine 受影響 Attestation/Relation，接著從 BeliefRevisionEvent + 仍有效的 evidence 重建 EpistemicState。不得只手動改目前 belief_level，否則無法 audit 或產 trajectory dataset。

6.19 Versioned Condition Schema Registry

每個 DomainPack MUST 註冊 condition JSON Schema 與比較語意；每筆 evidence/run 保存 conditions_schema_version。資料庫寫入時需驗證 schema。ConditionMatch 回傳 EXACT / COMPATIBLE / PARTIAL / INCOMPATIBLE / UNKNOWN，並可附 domain-defined tolerance 與 rationale。

6.20 Embedding Version & Re-index Policy

向量紀錄必須保存 embedding_model + embedding_version。Retrieval 不得直接混用不同 embedding space。換模型時使用 dual-index migration：新舊索引並行、背景 re-embed、驗證 recall/latency 後切換，再保留舊索引 rollback window。

6.21 Retraction / Erratum / Source-Work Dedup Gate

任何 external reported evidence 在導致重大 belief revision 前，SHOULD 檢查來源版本、撤稿/勘誤狀態與 source-work identity。Novelty Auditor 需保存 PriorArtSearchRecord，記錄查詢來源、query、時間窗與覆蓋限制；「沒搜到」只能支持 novelty candidate。

7. Cognitive / Multi-Agent Architecture - 科研認知功能分工

Laboratory Innovation Brain 的「多 Agent」是 cognitive decomposition，不是很多聊天機器人。Core 固定 6 個科研職責，DomainPack 可動態註冊 N 個 SpecialistRole；Model Router 提供 6 個 LLM logical slots (PRIMARY / ADVERSARIAL / FAST / CODE / PRIVATE_LOCAL / VISION) + 1 個 EMBEDDING slot。角色之間只透過有 schema 的 EvidenceBundle、Position、Hypothesis、CritiqueReport、VerificationPlan、RelationJudgment 與 Shared Scientific State 溝通。

圖 4. Cognitive architecture：Hypothesis Engine 彙整多個獨立 specialist position，Critic 攻擊 assumptions / confounders，Verification Planner 再把剩餘爭議轉成最低成本的 Evidence-Producing Action。

7.1 Core Cognitive Functions

Function	主要職責	能讀什麼	不能做什麼
Supervisor / PI	goal decomposition、 budget、priority、 termination、human gates	project state、role outputs、verification cost/capabilities	不能自行宣告物理真理 或把未驗證 idea 升級 成 verified
Evidence Researcher	找內外部 evidence、 condition match、 provenance、 conflict collection	papers、internal runs、episodes、 approved heuristics、external metadata	不能用自己生成文字補 證據
Hypothesis Engine	generate、 structure、dedup、 evolve competing hypotheses	evidence bundle、 open questions、 specialist positions	不能把 hypothesis 寫 回 evidence
Adversarial Critic	攻擊 assumptions、 confounders、causal gaps、 counterexamples、 falsifier	all admitted hypotheses + evidence summaries + specialist positions	不負責 final truth；不 能只靠語言說服力淘汰 假說
Verification Planner	把爭議轉成可觀察 predictions；選	hypothesis predictions、	不能預設一定要 simulation；不能直接

Function	主要職責	能讀什麼	不能做什麼
	literature / analytical / historical / numerical / surrogate / simulation / measurement 中最低成本充分驗證	available capabilities、cost、privacy、fidelity policy	執行任意 script
Novelty Auditor	外部 papers / patents / GitHub prior-art audit；區分 known / partially novel / novelty candidate	sanitized concept、public sources、approved metadata	預設不看 private raw data；沒搜到不能宣告全球唯一
Domain Specialist(s) - registered by DomainPack	依領域提供獨立機制判斷、conditions/validators、predictions	只讀該 DomainPack 授權的 evidence、rules、tools、historical cases	不是 core；不得直接改 core epistemic semantics 或 evaluator

7.2 Structured Scientific Debate Protocol

Default 3-stage protocol (stake-adaptive):
 Stage A - Independent positions from selected roles / specialists
 Stage B - Critic performs inverted retrieval + cross-examination
 Stage C - Surviving disagreement becomes explicit predictions + VerificationPlan

Escalate to deeper rounds only when:

- decision may REJECT a hypothesis,
- action is expensive/irreversible,
- evidence bundles materially conflict, or
- human requests deeper debate.

Structured Debate 的 anti-groupthink 來源不是「角色很多」，而是獨立 initial position、Critic 自己重跑 inverted retrieval、不同 EvidenceBundle 與 external evidence adjudication。系統 MUST 記錄 position diversity、bundle divergence、Critic 是否改變決策；簡單問題不得固定跑 8 rounds。當 `decision.stakes >= SourcePolicy.inverted_retrieval_threshold` 時，Critic MUST 執行 inverted retrieval，`CritiqueReport.inverted_bundle_id` 為條件式必填；未完成該步驟時 decision 不得進入 BELIEF_REVISION。

7.3 Model Routing - 6 個可拔 model slots + 1 embedding slot (不是 6 顆必備模型)

Logical Slot (6 LLM + 1 non-LLM)	主要用途	典型角色 / 工作
REASONING_PRIMARY	主要科研推理	Hypothesis 、 Semiconductor 、Optical 、 RF/System 、Verification Planner
REASONING_ADVERSARIAL	獨立反方與第二意見	Adversarial Critic / high- stakes independent critique path; provider can differ but evidence-path independence is primary
FAST_UTILITY	便宜快速的結構化工作	query rewrite 、分類、欄位抽 取、routing、輕量 rerank
CODE	程式與工具整合	parser 、typed-tool adapter 、 analysis code ；受 contract / test 約束
PRIVATE_LOCAL	敏感/離線任務	NDA 、未公開 idea 、air- gapped extraction / reasoning ；可映射 local model
EMBEDDING (non-LLM)	retrieval / similarity / proximity	documents 、evidence 、 case 、hypothesis dedup / clustering
VISION	多模態科學文件/圖表解析	figure/table/SEM/eye/field plot ；只在 multimodal ingestion 啟用時需要

7.4 Cognitive Role I/O Contract - 6 Core Roles + N Domain Specialists

核心固定 6 個 cognitive roles : Supervisor 、Evidence Researcher 、Hypothesis Engine 、
Adversarial Critic 、Verification Planner 、Novelty

Auditor 。Semiconductor/Optical/RF/Thermal 等由 DomainPack 註冊成 N 個 SpecialistRole ；

Heuristic Miner 屬於 knowledge-extraction service，不算核心科研角色。Model Router 可讓多角色共用模型，也可讓高 stakes critique 走獨立模型/獨立 evidence path。

Role	Input	Output	主要資料
Supervisor / PI	Human goal + current episode state	Research contract、 task order、budget/gates	project state、cost、 role summaries
Evidence Researcher	Structured question + missing evidence fields	Evidence Bundle + conflicts + source locators	internal artifacts/runs、 papers、episodes、 approved web
Hypothesis Engine	Evidence Bundle + open unknowns	2+ competing Hypothesis Certificates	normalized evidence、 mechanism graph、 prior hypotheses
Adversarial Critic	All positions + hypotheses + evidence map	attack report、 falsifiers、missing controls	same shared state; read-only on evidence
Verification Planner	Surviving hypotheses + disagreements + capabilities/cost	ranked VerificationPlan(s)	tool registry、 historical data、 fidelity/privacy policies
Novelty Auditor	Sanitized surviving concept	Prior-art matrix + novelty status	external paper/patent/GitHub /web only as allowed
Domain Specialist(s)	Question + selected claims/hypotheses + domain evidence	independent Position Object + predictions/confounders	DomainPack-specific evidence/rules/tools; N roles dynamically registered

7.5 Evidence Source Router - 資料選擇層怎麼選

Evidence Researcher 不把全部資料一次塞給 LLM。SourceRouter 先過 privacy/rights/ACL，再依 research intent 選 SourcePolicy，最後做 condition compatibility、source-work independence、authority/directness、semantic relevance、recency（按領域規則）與 retrieval cost。Critic 使用獨立 inverted policy 搜尋反例與替代機制。

Research intent	SourcePolicy	目的 / 防偏差
DIAGNOSIS	Internal matched runs / measurements / failure episodes first	快速找 root cause；仍需 contradiction check，避免只相信 lab history
MECHANISM_DISCOVERY	Internal evidence + peer-reviewed literature + contradictory evidence	比較 competing mechanisms；要求至少一條外部反證搜尋
NOVELTY_AUDIT	External literature + patents + GitHub/technical prior art	保存 PriorArtSearchRecord；不得用 internal novelty 代替 global novelty
CROSS_DOMAIN_INNOVATION	Adjacent-domain sources with exploration quota	避免歷史 bias；強制跨領域類比但 evidence type 分級
REPLICATION	Original work + independent replications + condition match	去除 preprint/journal/review 重複計數；優先可重現方法

SourcePolicy(intent):

DIAGNOSIS -> internal matched runs + historical failures + domain rules + external contradiction check

MECHANISM_DISCOVERY -> matched internal + peer-reviewed literature + contradictory literature + domain specialists

NOVELTY_AUDIT -> papers + patents + GitHub/technical prior art + explicit coverage record

CROSS_DOMAIN_INNOVATION -> force adjacent-domain exploration budget

REPLICATION -> original source + independent reproductions + method/condition comparability

Always: ACL/privacy -> condition/schema compatibility -> source-work dedup -> authority/directness -> relevance -> time/cost.

7.6 Model Independence & Inference Provenance

不同 provider 不是科學獨立性的充分條件。重大 REJECT / irreversible action 必須經 independent critique path：至少改變 retrieval bundle、reasoning policy 或 model route，並由 external evidence/verification adjudicate。所有 LLM 產物保存 InferenceProvenance；無 provenance 的舊推理不可作為新 belief transition 的依據。

8. Hypothesis & Epistemic Model - 可證偽假說與信念狀態

Laboratory Innovation Brain 必須把「idea」與「scientific hypothesis」分開。Idea 可以先自由生成；只有通過 Hypothesis Admission Gate，補齊 mechanism、prediction、falsifier、assumptions、confounders 與 minimum test，才可佔用昂貴 simulation budget。

8.1 Hypothesis Certificate

```
Hypothesis {
  hypothesis_id, statement, mechanism, assumptions[], predictions[],
  falsifier, confounders[], minimal_test_ref?,
  status_projection, belief_level_projection, parent_id?,
  created_in_episode, inference_provenance_id
}
```

Evidence for/against is resolved through provenanced RelationJudgments.
Current status is a projection of BeliefRevisionEvents, not the sole history.

Confidence 在 v1 不使用 0.58 這類未校準機率。建議先用 ordinal belief：LOW / MEDIUM / HIGH + support/contradiction counts + evidence quality dimensions。未來有足夠 benchmark 後，再導入 Bayesian posterior 或 calibrated score。

8.2 Hypothesis Lifecycle

```
DRAFT -> ADMITTED -> ACTIVE -> {SUPPORTED | CHALLENGED | CONTRADICTED | INCONCLUSIVE}
      \-> EVOLVED / SUPERSEDED
```

No LLM may directly assign a scientific transition.
Transition = TransitionPolicy.evaluate(current_state, admitted_relations, authority, condition_match, independence).
Every accepted transition emits BeliefRevisionEvent.

8.2.1 TransitionPolicy - belief 轉移不是 LLM 投票

TransitionPolicy 是版本化、可測試的 deterministic-first policy。LLM 可以產生 RelationJudgment 或解釋 rationale，但不得直接把 ACTIVE 改成 SUPPORTED/REJECTED。DomainPack 可擴充 authority/condition 規則；core 負責 event、policy version 與 projector。

```
TransitionPolicy {
  policy_id, version, domain?,
  from_state, candidate_to_state,
  required_relation_types[],
  required_authority_rule,
  required_condition_match[],
  min_independent_attestations?, independence_basis?,
  blocking_conflict_policy, human_gate?,
  effective_from, supersedes?
}
```

```
should_transition(hypothesis, relations, policy)
-> ALLOW | DENY | NEED_MORE_EVIDENCE | NEED_HUMAN_REVIEW
```

```
If any required AuthorityPolicy.compare() result is INCOMPARABLE:
-> NEED_HUMAN_REVIEW
-> auto-create ReviewItem(subject_type=AUTHORITY_CONFLICT,
```

subject_id=hypothesis_id, stakes=hypothesis.stakes)
-> no BeliefRevisionEvent may promote/reject until review resolves the authority relation.

8.3 Example - Ring Modulator Mechanism Competition

Hypothesis	Mechanism	Prediction	Falsifier
H1: Cj-dominant	reverse bias $\uparrow \rightarrow$ depletion width $\uparrow \rightarrow Cj \downarrow \rightarrow RC$ pole $\uparrow$	BW 改善與 $1/Cj$ 具有一致趨勢； 控制 R_s 後仍存在	Cj 幾乎不變而 BW 仍大幅改善
H2: R_s -dominant	depletion/access distribution 改變有效 series resistance	不同 bias 的 R_s 變化足以解釋 pole shift	固定/匹配 R_s 後改善仍保留
H3: electro-optic trade-off	electrical improvement 同時改變 carrier overlap / loss / detuning	system ER/BER 的最佳點不與 RC bandwidth 單調一致	跨 optical operating point 後系統表現仍完全由 RC 解釋

9. Active Verification Planning - 最低成本充分驗證與資訊增益

Optimizer 問「怎麼提高 FoM」；Scientific Verification Planner 問「目前哪一個最低成本的 Evidence-Producing Action，最能區分 competing mechanisms 或改變 scientific decision？」。這個 action 可以是查既有 evidence、解析計算、歷史案例、數值程式、surrogate、simulation、measurement 或 human review；不預設 simulation。

9.1 Least-Cost Sufficient Verification - v3.2 可計算定義

v1 不假裝擁有校準 posterior。Least-Cost Sufficient Verification 以明確 outcome space + TransitionPolicy 為基礎：候選 action 只有在至少一個「plausible outcome」能改變至少一個 ACTIVE hypothesis 的 policy decision，或解除 blocking conflict 時，才是 sufficient candidate。

```
plausible(outcome, action, domain_policy) =
  outcome is a member of declared OutcomeSpace
  AND outcome is not excluded by EXPLICIT constraints
  AND outcome respects backend/domain validity bounds
  AND outcome validity can be checked by ValidationReport-producing validators
```

```
sufficient(action) =
  exists plausible outcome y such that
  TransitionPolicy.evaluate(state | y) != TransitionPolicy.evaluate(state)
  OR y resolves a blocking provenance/condition/validity conflict
```

Selection:

1. remove non-sufficient actions
2. remove Pareto-dominated actions on CostVector
3. apply versioned deterministic SelectionPolicy fallback
4. record ranking rationale + policy version

Disagreement metrics are DomainPack/BenchmarkPolicy contracts: they MUST be deterministic, versioned and defined over the declared OutcomeSpace; core does not hard-code one universal distance.

若 prediction 無法數值化，DomainPack 必須宣告有限 ordinal outcome space，例如 $C_j = \{\text{DECREASE_STRONGLY}, \text{DECREASE_SLIGHTLY}, \text{UNCHANGED}, \text{INCREASE}\}$ 。Planner 不得自行想像「可能 outcome」來讓任意 action 變 sufficient。

9.2 v2+ : Information Gain / Value of Information

$$x^* = \operatorname{argmax}_x \mathbb{E}[H(\text{Belief_before}) - H(\text{Belief_after} \mid Y, x)] / \text{Cost}(x)$$

當 Laboratory Innovation Brain 累積足夠 hypothesis-outcome calibration data 後，再導入 Bayesian experimental design、ensemble posterior 或 surrogate uncertainty。不要在第一版假裝有機率模型。

9.3 Verification Cost / Capability Model

Verification Action	相對成本	何時用
Existing admitted evidence / memory	近乎 0	先重用現有 evidence、相似 episodes、已知 contradictions
Literature / patent / standard lookup	低	查已知機制、prior art、外部條件；受 rights/privacy gate
Analytical / rule / deterministic calculation	低	物理邊界、unit、trend、DRC、簡化模型快速排除
Historical-case / numerical analysis	低	相似 failure、已有 raw array 再分析、統計比較
Surrogate / reduced model	低-中	粗篩候選、預估敏感度；不可單獨終結核心 hypothesis
Low-fidelity simulation	中	發現趨勢與 provisional challenge
Standard / validation simulation	中-高	正式支持/反駁重要 mechanistic hypothesis
Physical measurement / fabrication / human expert gate	高-極高	sim-to-real、不可逆/昂貴決策、最終物理驗證

9.4 CostVector - 跨領域成本不是一個數字

Verification Planner 必須把成本視為向量，不把 6 小時模擬與 6 週 MPW shuttle 壓成一個不透明分數。預設欄位：wall_clock_s、human_minutes、money_estimate、compute_or_license_units、earliest_available_at、irreversible、dependency_risk。若需 scalar 排序，必須顯式記錄 policy/weights 並允許 Human override。

9.5 Capability Registry - Planner 先知道「能做什麼」才可能規劃

所有 evidence-producing backend 必須註冊 Capability descriptor。Planner 不直接看 provider 名稱，而看 requires/produces、availability、authority class、estimated cost、latency、privacy/license constraint。沒有 capability descriptor 的 backend 不可被自動規劃。

```
Capability {
  capability_id, domain?, action_type,
  requires[], produces[], conditions_schema_version?,
  authority_class, availability,
```

```

estimate_cost(params)->CostVector,
earliest_available_at?, irreversible,
privacy_constraints[], license_constraints[],
backend_id
}

```

9.6 Deterministic SelectionPolicy - Pareto front 不能留給 Agent 隨機挑

CostVector 先做 dominance filtering；若仍有多個 non-dominated candidates，MUST 使用版本化 SelectionPolicy。預設 fallback：先排除 irreversible（除非其 information necessity 為 REQUIRED），再依 earliest_available_at、human_minutes、wall_clock_s、money_estimate、license/compute units 的 lexicographic order；DomainPack/Project 可覆寫但必須保存 policy version。相同輸入、相同 policy 必須產生相同 ranked plan。

```

SelectionPolicy {
  policy_id, version,
  dominance_dimensions[],
  irreversible_rule,
  lexicographic_fallback[],
  human_override_allowed,
  tie_break_seed? = NONE
}

```

```
rank(actions, state, policy) -> ordered VerificationPlan
```

10. Verification Backend & Tool Architecture - Evidence-Producing Actions 與 Lumerical

Verification backend 在 Laboratory Innovation Brain 中不是「會想事情的 Agent」，而是產生或查證 evidence 的受控能力。Core 只透過 typed capability contract 呼叫 backend；Silicon Photonics 第一版的高價值 backend 是 Lumerical，但未來 domain 可以換成 DFT、training benchmark、統計分析、實驗機台或其他資料服務。

圖 5. Verification Ladder：從既有 evidence / analytical / historical data，逐層升級到 surrogate、simulation 與實體量測；升級只在較便宜層不足以改變判斷時發生。

10.1 Generic Evidence-Producing Backend Types

Verification Planner 對候選 action 統一看四件事：capability 是否存在、輸入條件是否滿足、expected evidence authority、cost/latency。Backend 可以不同，但輸出都必須回到 Artifact / Observation / Evidence / provenance contract。

Backend Type	典型用途	輸出
Existing Evidence / Memory	重用現有 run / measurement / episode	retrieved evidence bundle
Literature / External DB	已知機制、benchmark、prior art	cited evidence / metadata
Analytical / Rule Engine	公式、物理界限、unit/DRC checks	derived evidence + derivation
Numerical / Code Analysis	重算 curve、統計、plot、feature extraction	derived observation + artifact
Surrogate / Reduced Model	低成本篩選與敏感度	provisional simulated evidence
Simulator	高 fidelity domain verification	simulated evidence + validity metadata
Physical Experiment / Measurement	真實世界驗證	measured evidence + calibration/provenance
Human Expert Review	高風險、工具不足、tacit boundary	approved decision / heuristic; not automatic fact

10.2 Backend-bound run_* 與 Backend-agnostic extract_* 必須分離

Layer / Backend	Typed operation	Output / scientific role
CHARGE	run_charge_dc_sweep	bias sweep 、 carrier distribution 、 electrical state
CHARGE	run_charge_ac_sweep	AC current/impedance over frequency
Metric Extractor	extract_cj_per_length(array, conditions, basis)	可同時處理 CHARGE 或 measurement-derived impedance ； 輸出 method/normalization metadata
Metric Extractor	extract_rs_per_length(array, conditions, basis)	可同時處理 simulation/measurement ； 保留 de-embedding/geometry normalization
CHARGE	validate_contact_connectivity	access implant / contact continuity / floating regions
MODE	run_mode_solve	neff / mode profile / loss
Metric Extractor	extract_delta_neff(mode_data, conditions)	不綁 MODE product name ； 輸出 extraction method/provenance
Metric Extractor	extract_overlap_metrics(field_data, conditions)	backend-agnostic metric extraction
FDTD	run_coupling_simulation	coupler / bend / scattering / field distribution
INTERCONNECT	run_ring_spectrum	resonance / Q / FSR / ER / IL
Workflow/Backend	run_interconnect_nrz_workflow	drive/receiver/DSP/decision assumptions 全部進 manifest ； 不是單一原子 tool
Workflow/Backend	run_thermal_drift_lock_workflow	thermal/controller assumptions 進 manifest ； 輸出 typed arrays + run provenance

run_* 只負責執行 backend 並產生 immutable artifacts / typed numerical outputs ; extract_* 只接受 typed data + conditions + normalization basis 。這讓同一個 Cj/Rs/Q/ER 定義可套用 simulation 與 measurement ，避免 sim-to-real 其實是在比較兩套不同 extractor 。

10.3 Tool Contract Requirements

- Input schema : units 、 allowed ranges 、 required project/artifact references 。
- Immutable manifest : solver version 、 material database 、 mesh 、 boundary conditions 、 script commit 。
- Preflight validators : geometry / connectivity / DRC / unit consistency 。
- Run isolation : 每一次 run 有獨立 working directory / run_id / logs 。
- Output schema : raw artifact URI + extracted metrics + warnings + convergence status 。
- Evaluator integrity : Agent 不可直接修改 metric extractor 或 hidden validation config 。
- Replay : 同一 manifest 能重跑；若版本不同，明確標為 non-identical reproduction 。

10.4 Simulator Exploitation Defense

- DEV evaluator : 允許 Agent 看 coarse / public metrics 。
- Validation gate : 使用不同 mesh 、 hidden bias points 、 process variations 、 alternative numerical settings 。
- Final test : full multiphysics / independent rerun / measurement ；具體規則可對 Agent 隱藏 。
- 不可只看單一 scalar FoM ；至少保留 Pareto vector 與 constraint violations 。

10.5 EvidenceAuthority - core 不假設 measurement 永遠最高

EvidenceAuthority 不採固定「measurement > simulator」階梯。任何 backend result 的 authority 由 method 、 calibration/convergence 、 validated-range fit 、 condition match 、 independent replication 等共同決定。Silicon Photonics 可用 mesh/solver/calibration validity ；其他 domain 可提供自己的 validity schema 。低 authority contradiction 只能觸發 PROVISIONAL_CHALLENGE ，不可單獨 REJECT 高 stakes hypothesis 。

Exploratory Fidelity

- analytical estimate / surrogate / coarse mesh
- 可：排序候選、發現異常、提出 provisional challenge 、淘汰明顯 violation
- 不可：單獨將核心 hypothesis 標為 REJECTED

Standard Fidelity

- laboratory-approved nominal mesh/model/tolerance
- 可：形成正式 SUPPORT / CONTRADICT evidence ；仍需保存 convergence/validation metadata

Validation Fidelity

- mesh sensitivity / independent rerun / hidden bias-process corners / alternative numerical settings

- 可：將 provisional contradiction 升級為 CONFIRMED_CONTRADICTION；支援重大 belief revision

Physical Measurement

- 進入 sim-to-real adjudication；量測也須帶 calibration、uncertainty、instrument/procedure provenance。

因此 contradiction 至少分 PROVISIONAL_CHALLENGE 與 CONFIRMED_CONTRADICTION。

當 low-fidelity 結果與 hypothesis 不一致，預設動作是觸發 fidelity escalation / diagnostic check，而不是直接砍掉該 hypothesis。

10.5.1 AuthorityPolicy Comparator - belief gate 的核心運算子

EvidenceAuthority 是偏序（partial order），不是固定 measurement > simulation 全序。

DomainPack 註冊 AuthorityPolicy；若兩份 evidence 在不同有效範圍/方法上不可比較，必須回傳 INCOMPARABLE，不能硬排大小。

AuthorityComparison = STRONGER | WEAKER | EQUIVALENT | INCOMPARABLE

```
AuthorityPolicy {
  policy_id, version, domain?,
  compare(a: EvidenceAuthority, b: EvidenceAuthority) -> AuthorityComparison
  meets(required_rule, candidate) -> bool
}
```

Comparison dimensions may include:

method validity, calibration/convergence, validated-range fit,
ConditionMatch, independence/replication, known limitations.

10.6 Backend Validity Metadata - Simulator 只是其中一種 Evidence Producer

Core 只要求 backend_validity 與 EvidenceAuthority；Silicon Photonics DomainPack 可定義 simulator-specific validity（solver/mesh/material/calibration），measurement backend 可定義 calibration/de-embedding/instrument validity。重大 contradiction 必須滿足 policy 所需 authority。

```
EvidenceAuthority {
  authority_class, method_type, condition_match,
  validity_status, validated_range_fit, independent_replication_status?,
  limitations[], policy_version
}
```

```
BackendValidity {
  backend_id, schema_version, validity_payload{},
  warnings[], validation_artifact_refs[]
}
```

Silicon Photonics may register SimulatorValidityV1 inside its DomainPack.

如果 simulator 與 measurement 衝突，Epistemic State 建立 SIM_TO_REAL_CONFLICT。系統應提出可能來源：模型近似、process variation、measurement calibration、parameter mismatch、boundary/mesh、material model 等，並選擇能區分它們的下一步，不可用「measurement 永遠真」或「simulator 已校準所以一定真」的硬規則抹平。

10.7 Job Queue / License-seat / Retry Contract

長時 simulator、measurement、fabrication 不可被同步 state machine 阻塞。ToolExecutor 先建立 Job，資源可用時執行；license-seat 不足、rate limit 或 device unavailable 必須進 retry/backoff/blocked state，而不是讓 LLM 重複呼叫。每個 job 具有 idempotency key、attempt count、deadline/timeout policy、trace_id 與 partial-artifact policy。

11. Silicon Photonics Domain Pack - 第一個垂直領域

Laboratory Brain core 不知道「Cj 是什麼」；Silicon Photonics Domain Pack 定義 domain entities、physics validators、metric extractors、condition schema、specialist prompts 與 verification backends。Lumerical 是這個 DomainPack 的 simulator backend，而 literature / analytical / historical / numerical / measurement 也必須能被同一 Verification Planner 選用。

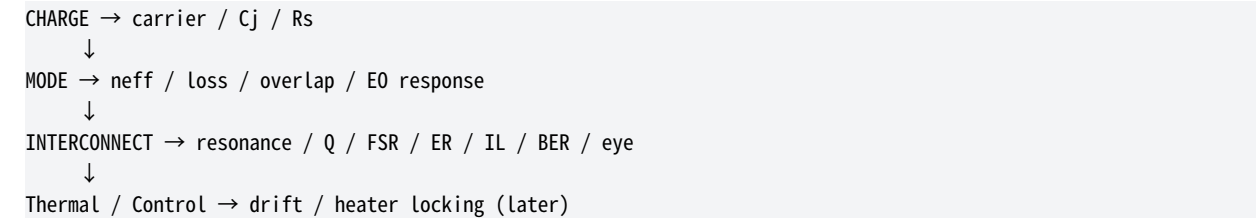
11.0 Silicon Photonics Verification Backends

v3.2 第一個 DomainPack 必須至少註冊：internal evidence/case retrieval、literature adapter、analytical/rule checks、Python numerical analysis、Lumerical typed tools；measurement adapter 可先保留 contract。這樣同一個 research question 不會被強迫全部轉成 Lumerical jobs。

11.1 Domain Cognitive Modules

模組	主要概念	v1 / future
Semiconductor	PN junction、depletion、carrier transport、doping、Cj、Rs、contact、mesh	v1 core
Optical	mode、neff、loss、Q、FSR、coupling、resonance	v1 core
RF/System	RC bandwidth、50Ω load、electrode、driver、ER/IL/BER/eye	v1 core
Thermal/Control	thermo-optic drift、heater、feedback locking	v1.5 / v2
Fabrication	PDK / DRC / process corners / variability	v2
Communication System	WDM / link budget / channel rule / DSP assumptions	v2

11.2 Ring Modulator First Vertical



這條鏈的優點是每一層都有不同 confounder，可測試 Laboratory Innovation Brain 是否真的能跨 domain 整合 evidence，而不是只做單一 optimizer。

11.3 Domain Evidence Conditions

- SOI stack：core/slab thickness、BOX/TOX、material model version。

- Device geometry : ring radius 、 coupling gap 、 waveguide width 、 segment length °
- Junction : topology 、 implant geometry 、 doping concentration 、 contact/access connectivity °
- Electrical : DC bias 、 AC frequency 、 driver/load 、 temperature °
- Optical : wavelength 、 polarization 、 laser detuning 、 power °
- System : bit rate 、 modulation format 、 receiver settings 、 eye analyzer assumptions °

12. End-to-End Verification Loop - 從問題到具驗證等級的創新候選

圖 6. Laboratory Innovation Brain 研究閉環：Evidence 與 Hypothesis 交替演化；Verification Planner 只在較便宜證據不足時升級到 simulator / physical experiment。

12.1 Event-driven State Machine - 支援等待數小時到數週的科工作

```
NEW → STRUCTURED → EVIDENCE_GATHERING → HYPOTHESIZING → CRITIQUING
→ VERIFICATION_PLANNING
→ [AWAITING_JOB | AWAITING_HUMAN | AWAITING_EXTERNAL_SOURCE]
→ EVIDENCE_ADMISSION → BELIEF_REVISION → NOVELTY_AUDIT → DECISION
→ CLOSED / DEFERRED / BLOCKED

Resume trigger = JobCompleted / HumanDecision / ExternalSourceAvailable / BudgetUpdated event.
```

12.2 Innovation Candidate Output Contract

系統最後不是輸出「已證明為真且全球唯一的創新」，而是輸出一份可追溯 Innovation Candidate Certificate，將 scientific verification 與 novelty 分成兩條獨立軸。這允許系統誠實表達「很新但尚未驗證」或「物理很穩但已是 known work」。

Axis	State	意義
Verification	L0 IDEA	只有結構化概念 / hypothesis，尚無外部支持
Verification	L1 EVIDENCE-GROUNDED	已有可追溯 literature/internal evidence
Verification	L2 ANALYTICALLY / COMPUTATIONALLY VERIFIED	解析、數值或歷史資料已驗證核心 prediction
Verification	L3 SIMULATION / CROSS-VERIFIED	標準/validation fidelity 或多方法一致
Verification	L4 EXPERIMENTALLY VERIFIED	已有實體量測 / fabrication evidence
Novelty	KNOWN / PARTIALLY NOVEL / NOVELTY CANDIDATE / STRONGLY SUPPORTED NOVEL	與外部 prior art 分開評估；「沒搜到」只能是 candidate

12.3 Termination Conditions

- Budget exhausted : token / solver / wall-clock / experiment budget 。
- Hypothesis convergence : 剩餘 hypotheses 對可行實驗的 predictions 幾乎無法區分 。
- Evidence saturation : 新 retrieval 不再改變 support/contradiction structure 。
- Goal reached : 達到 design constraint / mechanism identification / anomaly diagnosis acceptance criteria 。
- Human gate : 需要 fabrication / NDA external query / irreversible action 。
- Uncertainty escalation : 系統判定目前工具不足，標記 open question 而非硬猜 。

12.4 Example Episode - Rs 異常診斷

範例：CHARGE 中 $C_j(V)$ 合理下降，但 R_s 異常偏高且幾乎不隨 reverse bias 變化。Laboratory Innovation Brain 應先從 internal case memory、文獻與 semiconductor physics 產生至少三個 competing causes : contact/access discontinuity、mesh/convergence、material/contact model / normalization。然後選擇「最便宜且最能區分」的檢查：geometry connectivity validator + mesh sensitivity，而不是立刻掃 50 組 doping 。

```

Observation 01:  $C_j \downarrow$  with reverse bias;  $R_s \approx$  flat and very high
H1: implant-to-contact gap / floating access path
H2: access/junction mesh insufficient
H3: extraction / normalization bug
H4: physical  $R_s$  dominated by another fixed path

Test 1: connectivity graph + current-density path check
Test 2: mesh refinement only
Test 3: independent impedance extraction
→ Evidence separates implementation artifact from physical mechanism

```

12.5 Execution Trace - 科學 provenance 與執行 provenance 都要完整

每個 episode 建立 trace_id，向下串 round/LLM call/retrieval/tool job/run/artifact 。

Hypothesis、Critique、Relation Judgment 與 BeliefRevisionEvent 都保存 inference/execution provenance。這讓「哪個模型看哪包資料，因哪個結果把 H1 改成 CHALLENGED」可完整重播。

13. Long-term Learning - 讓實驗室大腦隨時間變強

Laboratory Innovation Brain 的「學習」不應一開始就等於 fine-tune。最先產生價值的是可檢索的高品質 research episodes 與 failure cases。Weights 應學 procedural/research policy；快速更新的 papers、PDK、measurement 仍留在外部 memory。

13.1 Learning Ladder

階段	方法	學到什麼	風險
L0	RAG + structured evidence	知道最新/內部 facts	retrieval quality
L1	Case-based reasoning	遇到新 anomaly 能找相似 episode / failure	case representation quality
L2	Hypothesis pattern mining	常見 mechanism / confounder / test template	可能強化歷史偏見
L3	Trajectory distillation / SFT	Evidence→Hypothesis→Experiment→Revision 的 research policy	data leakage / teacher bias
L4	Preference learning	偏好較可證偽、較省成本、較物理合理的 proposal	preference target 設錯
L5	RL / offline RL from evaluators	在 bounded design/search tasks 學 action policy	reward hacking / simulator exploitation
L6	Continual domain adaptation	domain terminology / procedural competence	catastrophic forgetting / data governance

13.2 三種 Memory vs Weights

類型	內容	是否進 weights
Semantic / explicit	papers、PDK、最新數據	通常不；RAG / DB
Episodic	我們試過什麼、結果、失敗	先 case memory；可蒸餾成 trajectories
Procedural / policy	如何診斷、如何設 experiment、如何選 fidelity	最值得後期 SFT / preference / RL

13.3 不建議即時 Online Gradient Update

一次 simulation 的結果可能是 solver failure、bad mesh、錯誤 extraction 或真正 scientific negative result。若立即更新 weights，很容易把 artifact 學成物理定律。v1-v2 建議：每輪只更新 structured memory；訓練採批次、審核、版本化、可回滾。

14. Security, Privacy & Governance - 私有/混合/連網治理

圖 7. 三種資料使用模式：Private、Research、Novelty Audit 都先經 Local Privacy Gateway。

14.1 Sensitivity Labels

標籤	例子	外部使用
RESTRICTED_NDA	foundry PDK、NDA docs、private process rules	禁止外部 API / cloud model
CONFIDENTIAL_LAB	unpublished topology、measurement、patentable idea	預設 local；需 explicit project policy
INTERNAL	meeting、thesis draft、internal report	可依 policy 使用內部/approved cloud processing
PUBLIC	published paper、public patent、open-source repo	可外部搜尋與雲端模型

14.2 Modes

Mode	資料來源	外部連線	用途
Private Mode	local artifacts / local model / local tools	No egress	NDA / unpublished idea / sensitive analysis
Research Mode	local evidence + approved external literature	sanitized query / approved model	一般研究與文獻補充
Novelty Audit	sanitized hypothesis summary + public prior-art search	papers / patents / GitHub / web	檢查 open-world novelty；與內部 reasoning 分離

14.3 Governance Gates

- 外部查詢 gate：private identifiers / exact confidential geometry 不得送出。
- 昂貴計算 gate：超出 session solver budget 需 supervisor/human approval。
- Fabrication gate：版圖/製程送件必須由 human sign-off。
- Memory admission gate：任何 inferred claim 標為 inferred；measurement/simulation 需 artifact reference。
- Deletion/governance：科研資料預設 append-only / supersede；仍保留依法律、NDA、個資治理真正刪除的能力。

14.4 Actor / ACL / Review Queue - 沒有「誰」就沒有正 governance

最小 Actor 模型包含 human、service account、agent role；ACL 至少支援 project membership、sensitivity clearance、approval authority。所有 external egress、private repo access、human approval 與 destructive governance action 都要記 actor_id。Human ReviewQueue 必須有 stakes、created_at、SLA/expiry policy 與 queue capacity，避免所有 uncertain item 永久卡在 PENDING。

14.4.1 ReviewQueue 必須接回 Verification Planner

Human review 是一個 Capability，不是無限免費資源。ReviewQueue 的 queue depth、reviewer availability、SLA 與 expiry 會更新 human-review Capability 的 availability / earliest_available_at / estimated human_minutes；Planner 因此會把教授/研究員的注意力當成真正成本。

```
ReviewItem {
  review_id, project_id, episode_id, trace_id,
  subject_type, subject_id, stakes, reason,
  required_authority?, required_role?,
  created_at, due_at?, expires_at?,
  status: QUEUED|ASSIGNED|APPROVED|CORRECTED|REJECTED|EXPIRED,
  assigned_actor_id?, decision_ref?,
  estimated_human_minutes
}
```

```
HumanReviewCapability.availability <- ReviewQueue capacity + actor schedule
```

14.5 Secret / License Ingestion Guard

Watcher 在 immutable ingest 前先做 secret scanning；疑似 credential 不得直接永久封存，需 quarantine/redaction policy。External code 保存 license_class = PERMISSIVE / COPYLEFT / PROPRIETARY / UNKNOWN；UNKNOWN/COPYLEFT 原始碼不得默認進入 code-generation context，除非 project policy 明確允許。

15. Evaluation & Benchmark Suite - 如何證明的會究

Laboratory Innovation Brain 成功不能只看「LLM judge 覺得 idea 有創意」。評估要分科研正確性、創新性、效率、可追溯性、長期學習與人類實用性。

15.1 Benchmark Classes

Benchmark	任務	Ground truth / 評估
Anomaly Diagnosis	由異常 observation 找 root cause + minimal verification	已知故障注入 / 歷史真實 case
Mechanism Discrimination	多 competing mechanisms，選下一個 test	synthetic/analytical simulator + known causal model
Hypothesis Quality	從 evidence 提出可證偽 hypothesis	expert rubric + prediction/falsifier completeness + evidence grounding
Experiment Efficiency	在有限 solver budget 下得到正確 conclusion	number of calls / cost / regret / information gained
Design Discovery	找 Pareto-improved design 並說明 mechanism	baseline optimizer comparison + robustness
Cross-paper Synthesis	跨 subfield evidence 形成 non-obvious connection	held-out known discovery / expert review
Novelty Audit	判斷 internal idea 是否有 prior art	date-cutoff literature / patent corpus
Sim-to-Real	用 measurement 修正 simulator-based belief	held-out fabrication/measurement result
Longitudinal Learning	第 N 次遇到相似問題是否更快	episode history ablation / time-split benchmark

15.2 Core Metrics

面向	Metric examples
Validity	physics constraint violations、root-cause accuracy、prediction calibration
Evidence grounding	citation/locator correctness、condition match、unsupported claim rate
Falsifiability	有明確 falsifier 的 hypothesis 比例、prediction executability
Efficiency	solver calls、wall time、cost-to-correct-conclusion、invalid run rate
Innovation	prior-art distance、non-obvious cross-domain connection、expert novelty rating
Mechanistic insight	能否提出 transferable mechanism，不只 design-specific correlation
Robustness	process corners / hidden conditions / independent rerun

面向	Metric examples
	performance
Memory value	similar-case retrieval hit rate 、 repeated-failure reduction
Human usefulness	PI/student acceptance 、 time saved 、 decision quality 、 explanation usefulness

15.3 Required Ablations

- A. Strong single LLM
- B. + RAG
- C. + Structured Evidence
- D. + Hypothesis Certificate / Critic
- E. + Multi-specialist reasoning
- F. + Active Experiment Selection
- G. + Episodic / Failure Memory
- H. Full Laboratory Innovation Brain

這樣可以回答 reviewer /教授最重要的問題：「到底是多 call 幾次模型比較強，還是哪個 architecture mechanism 真正有用？」

15.4 Structured Debate / Retrieval Independence Metrics

Structured Debate 是否減少 groupthink 必須可證偽。最低指標：Round/Stage A position pairwise semantic diversity、Critic evidence-bundle divergence、Critic 改變最終 hypothesis set 的比例、surviving-hypothesis diversity、additional evidence cost。若增加多 Agent 只增加 token 而不改善這些指標，應縮減角色/round。

16. Roadmap - 從基礎建設到真實實驗室閉環

圖 8. 建議演進路線：先做 provenance 與 typed tools，再做 research memory / hypothesis / verification loop，最後才是長期學習與 sim-to-real。

Narrative Stage	對應 Milestone	說明（非獨立 gate）
Foundation	M0a + M0b	資料 identity/schema/CI 與 actor/budget/event/trace 地基；正式 Exit Gate 只看 § 26.1。
Research Memory	M1	Ingestion、condition/source policy、provenance、async job。
Silicon Photonics Tools	M2	DomainPack、Capability、run/extract、backend validity。
Hypothesis Brain	M3	6 core roles + specialists、debate、inverted retrieval。
First Vertical	M4	VS-SP-001 end-to-end。
External Evidence Expansion	DEFERRED M5	GitHub/Literature/Patent/Web 等 connector 擴張。
Long-term / Real Lab	DEFERRED M6+	Multi-physics、operational lab、learning、second domain。

16.1 可衍生研究題目（副產品，不限制 Laboratory Innovation Brain）

- Evidence-grounded hypothesis competition for silicon photonics。
- Active discriminative multiphysics verification action selection。
- Condition-aware scientific RAG / evidence normalization for photonic devices。
- Failure-memory / case-based diagnosis of photonic simulations。
- Simulator-grounded belief revision vs LLM self-critique。
- Longitudinal learning of a laboratory research agent from experiment trajectories。

- Sim-to-real scientific belief update for photonic device modeling °

17. Implementation Contracts & Schemas - 核心資料結構

本章提供 v3.2 可直接用 Pydantic / SQLAlchemy / event-store pattern 實作的概念 schema。Claim identity、InferenceProvenance、BeliefRevisionEvent、Actor、Job、Capability、CostLedger、Condition schema version 與 trace_id 為新增硬契約。

17.1 Artifact Schema

```
Artifact {
  artifact_id, content_hash, media_type, uri,
  lineage_id?, lineage_revision?, previous_artifact_id?,
  source_origin, sensitivity_label, project_id,
  created_at, captured_at, author_or_device?, actor_id?,
  parser_version?, derived_from_artifact_ids[],
  rights_metadata?, secret_scan_status, metadata{}
```

artifact_id is content-addressed identity.

Changing 1 byte creates a new artifact_id/hash.

lineage_id + lineage_revision express human/version lineage when needed.

17.2 Claim / Observation / Attestation (Evidence) Schema

```
Claim {
  claim_id, normalized_proposition, domain?, scope{},
  identity_status, created_at
}
```

```
Observation {
  observation_id, run_id?/artifact_id, metric_or_event, value_ref?,
  conditions{}, conditions_schema_version, method_ref, created_at
}
```

```
Attestation {
  attestation_id, claim_id?/observation_id?, epistemic_type,
  source_artifact_id?/source_work_id?/run_id?, locator,
  conditions{}, conditions_schema_version, field_states{},
  units?, uncertainty?, method{}, extraction_status, verification_status,
  authority_class?, created_at, extractor_version, extraction_provenance
}
```

No support_targets[] / contradict_targets[] here. Those are RelationJudgments.

17.2.1 Observation / Attestation Creation Rule

Internal backend outputs first create Observation linked directly to Run/Artifact;它不需要再建立一筆「自我 Attestation」才能存在。Attestation 主要表示外部/報告型來源對 Claim 或 Observation 的具名見證。若 internal result 被寫入 paper/report，再建立 SourceWork/Attestation 表示「該文件如何報告它」，避免把 observation 與報告它的文件混成同一物件。

17.3 Research Episode Schema

```
ResearchEpisode {
  episode_id, project_id, goal, research_contract_id, trace_id,
  parent_episode_id?, state, start_time, end_time?,
  hypothesis_ids[], verification_plan_ids[], job_ids[], run_ids[],
  decision_ids[], outcome_status, failure_analysis_id?,
  human_annotations[], cost_ledger_id?
}
```

17.4 Run Manifest Schema

```
Run {
  run_id, job_id, capability_id, backend_id, domain?, trace_id,
  input_artifacts[], input_parameters{}, conditions{}, conditions_schema_version,
  environment{}, code_provenance, backend_validity{},
  status, warnings[], output_artifacts[], numerical_array_refs[],
  start_time, end_time, reproducibility_manifest_hash
}
```

Core MUST NOT hard-code solver_settings / mesh_convergence / calibration fields.
Domain/backend validity schema owns those semantics.

17.5 Hypothesis Schema

```
Hypothesis {
  hypothesis_id, statement, mechanism, assumptions[], predictions[],
  falsifier, confounders[], minimal_test_ref?, parent_id?,
  status_projection, belief_level_projection,
  created_in_episode, inference_provenance_id
}
```

No evidence_for[] / evidence_against[] / triggering_evidence_ids[] arrays.
Support/contradiction is resolved only through RelationJudgment.
Status is rebuilt from BeliefRevisionEvent + TransitionPolicy.

17.6 Failure Analysis Schema

```
FailureAnalysis {
  failure_analysis_id, episode_id,
  symptom, expected_behavior, observed_behavior,
  candidate_causes[], confirmed_root_cause?,
  root_cause_evidence_ids[], failure_class,
  fix?, prevention_rule?, resolution_status
}
```

17.7 Expert Heuristic Schema

```
Heuristic {
  heuristic_id, trigger_pattern, context_constraints{},
  suggested_checks[], rationale, source_person_or_episode,
  source_artifact_ids[], derived_from_failures[], validated_on_episodes[],
  status, approval_status, approved_by_actor_id?, approved_at?,
  last_reviewed_at
}
```

17.8 Graph Edge Schema

```
RelationJudgment {
  relation_id, from_entity_id, to_entity_id, relation_type,
  attributes{}, supporting_attestation_ids[],
  condition_match_ref?, inference_provenance_id?,
}
```

```
actor_id?, valid_from, valid_to?, created_at
}
```

Core relations: SUPPORTS, CONTRADICTS, TESTS, PREDICTS, PRODUCES, DERIVED_FROM, INSTANTIATES, SUPERSEDES, SAME_WORK_AS, CITES, SUGGESTS_CHECK.

Relation is the single source of truth for support/contradict semantics.

17.9 Evidence Field Status Contract

每個重要條件/參數可獨立持有 field status，而不是整個 Evidence 只有一個 confidence。推薦枚舉：EXPLICIT、DERIVED、INFERRED、UNKNOWN、NOT_REPORTED、REFERENCED_EXTERNALLY、CONFLICTING。只有 EXPLICIT / VERIFIED_DERIVED 可直接作 condition filtering 的高權重依據；INFERRED 必須被標記為非原始 evidence。

```
EvidenceField { name, value?, unit?, status, locator?, derivation?, source_refs[], reviewer_status? }
```

17.10 EvidenceAuthority / BackendValidity Contract

重大 belief revision 所用 evidence 必須帶 EvidenceAuthority + BackendValidity；simulator-specific validity 由 DomainPack/backend schema 提供。

Silicon Photonics SimulatorValidityV1 is a DomainPack-specific BackendValidity schema; core has no standalone SimulatorValidity table.

17.11 Candidate Heuristic Certificate

Heuristic Miner 的輸出不是直接入庫的 active rule，而是候選憑證。它必須引用來源 artifact/episode、保留原句 locator，並等待人類 approval。

```
CandidateHeuristic {
  candidate_id, trigger_pattern, suggested_checks[], rationale,
  source_artifact_ids[], source_locators[], source_episode_ids[],
  miner_model_version, status=PENDING_REVIEW,
  proposed_scope{}, conflicts_with_existing_rules[]
}
```

17.12 GraphRepository Interface Contract

上層 cognition/retrieval 只依賴 GraphRepository contract；v1 backend 為 PostgreSQL。任何未來 graph backend 必須先通過 relation semantics、provenance trace、transaction consistency 與 regression tests，再允許切換。

```
GraphRepository {
  neighbors(entity_id, relation_types?, as_of?, limit, cursor?)
  find_paths(start, goal?, relation_types?, max_depth, as_of?, limit)
  trace_provenance(entity_id, as_of?, limit)
  find_support_chain(target_id, as_of?, limit)
  find_contradictions(target_id, as_of?, limit)
  find_related_failures(query, limit)
  upsert_relation(...)
  invalidate_relation(relation_id, reason, actor_id)
  batch_upsert(...)
}
```

All traversal APIs MUST be bounded; as_of semantics are required for historical replay.

17.13 BeliefRevisionEvent & EpistemicState Projection

Belief revision 採 append-only event；EpistemicState projector 可由全量 event replay 或 checkpoint + delta 重建。任何 manual correction 也必須形成 event，不可直接 UPDATE current belief row。

```
BeliefRevisionEvent {
  event_id, target_type, target_id,
  from_state?, to_state,
  triggering_attestation_ids[], triggering_relation_ids[],
  policy_version, actor_id?, inference_provenance_id?,
  rationale_artifact_or_record_ref?, occurred_at, trace_id
}

EpistemicStateProjection {
  target_id, current_state, belief_level?, unresolved_conflicts[],
  last_event_id, projection_version, projected_at
}
```

17.14 InferenceProvenance Contract

所有 LLM 生成/判斷物件必填。evidence bundle hash 讓同一 prompt/model 但不同 retrieval context 可區分。

```
InferenceProvenance {
  inference_id, role, logical_slot,
  provider?, model_id, model_version,
  prompt_id, prompt_version,
  evidence_bundle_hash, source_policy_version?,
  parameters{temperature?, seed?, reasoning_mode?},
  created_at, trace_id
}
```

17.14.1 EvidenceBundle / Debate Objects / VerificationPlan Contracts

EvidenceBundle 是 retrieval 的可重現封裝，也是所有 LLM 科研推理 provenance 的輸入 identity。Canonical hash 使用 deterministic serialization：固定 schema version、排序後 attestation IDs、query/policy/condition filters 與 source snapshot refs；不得 hash 任意 JSON 字串順序。

```
EvidenceBundle {
  bundle_id, schema_version, research_intent,
  query_text, query_hash, source_policy_id, source_policy_version,
  condition_filter{}, condition_schema_versions{},
  ordered_attestation_ids[], source_snapshot_refs[],
  retrieval_trace_id, created_at, canonical_hash
}
```

canonical_hash(bundle) = SHA256(JCS/RFC8785 canonical JSON of hash_fields)

```
Position {
  position_id, role_id, episode_id, hypothesis_refs[],
  mechanism_view, supporting_relation_ids[], uncertainties[],
  predictions[], confounders[], bundle_id, inference_provenance_id
}
```

```
CritiqueReport {
  critique_id, target_ids[], objections[], alternative_mechanisms[],
```

```

    falsifier_challenges[], blocking_conflicts[],
    primary_bundle_id, inverted_bundle_id,
    inference_provenance_id, created_at
}

VerificationPlan {
    plan_id, episode_id, candidate_action_ids[], ranked_action_ids[],
    sufficiency_results{}, pareto_front_ids[],
    selection_policy_id, selection_policy_version,
    chosen_action_id?, rationale_ref, created_at
}

ResearchContract {
    contract_id, project_id, question, intent, success_criteria[],
    constraints{}, privacy_mode, budget_id?, stop_conditions[], actor_id
}

Decision {
    decision_id, episode_id, decision_type, subject_id,
    result, policy_refs[], triggering_event_ids[], actor_id?, created_at
}

```

17.15 Actor / ACL Contract

```

Actor { actor_id, actor_type:HUMAN|SERVICE|AGENT_ROLE, display_name?, active }
ProjectMembership { actor_id, project_id, role, sensitivity_clearance[], approval_scopes[] }

```

Default rule: new internal artifact sensitivity is fail-closed; explicit authorized action is required to lower classification.

17.16 Job / Suspend-Resume Contract

```

Job {
    job_id, episode_id, capability_id, trace_id, idempotency_key,
    state:QUEUED|RUNNING|WAITING_RESOURCE|SUCCEEDED|FAILED|CANCELLED,
    attempt_count, max_attempts, submitted_at, started_at?, finished_at?,
    retry_policy{}, timeout_policy{}, resource_requirements{},
    result_run_id?, structured_error?
}

```

17.17 CostLedger & Budget Gate

```

CostEntry {
    cost_entry_id, episode_id, actor_or_slot, action_ref,
    tokens_in?, tokens_out?, wall_clock_s?, human_minutes?,
    money_estimate?, compute_units?, license_seat_s?,
    earliest_available_at?, irreversible?, recorded_at
}

```

Before each LLM/tool call, BudgetGate MUST check project/episode caps. Exceed -> refuse/escalate; never silently continue.

17.18 Capability Descriptor

```

Capability {
    capability_id, domain?, action_type, backend_id,
    requires[], produces[], authority_class, availability,
    conditions_schema_version?, privacy_constraints[], license_constraints[],
    irreversible, earliest_available_at?,
    estimate_cost_contract, version
}

```

17.19 Condition Schema / ConditionMatch

```
ConditionSchemaRegistration { domain, schema_id, version, json_schema_ref, comparator_version }
ConditionMatch { state:EXACT|COMPATIBLE|PARTIAL|INCOMPATIBLE|UNKNOWN, matched_fields[], mismatches[], unknowns[],
tolerance_policy_version, rationale_ref? }
```

17.19.1 Budget / ReviewItem / Observability Contract

BudgetGate 需要明確 scope 與可調整權限；ReviewQueue/observability 也必須是可測試 contract，而不是只有 prose。

```
Budget {
  budget_id, project_id, episode_id?,
  hard_caps{}, soft_caps{}, currency?,
  effective_from, expires_at?, approved_by_actor_id, version
}

ReviewItem {
  review_id, project_id, episode_id, trace_id,
  subject_type, subject_id, stakes, reason,
  required_authority?, required_role?,
  created_at, due_at?, expires_at?,
  status:QUEUED|ASSIGNED|APPROVED|CORRECTED|REJECTED|EXPIRED,
  assigned_actor_id?, decision_ref?, estimated_human_minutes
}

ExecutionSpan {
  span_id, trace_id, parent_span_id?, span_type,
  episode_id?, actor_id?, model_call_id?/job_id?/retrieval_id?,
  start_time, end_time?, status, cost_entry_ids[], metadata{}
```

17.19.2 BenchmarkPolicy / OutcomeSpace / ValidationReport

```
BenchmarkPolicy {
  policy_id, domain, benchmark_set_id,
  metric_key, threshold, direction?,
  calibrated_at, sample_size,
  calibration_artifact_refs[], version, active
}

OutcomeSpace {
  outcome_space_id, domain, action_type, hypothesis_type?,
  schema_version, outcomes[], order_or_metric_ref?,
  explicit_exclusions[], validity_bounds{}, version
}

ValidationReport {
  report_id, subject_type, subject_id,
  validator_id, validator_version,
  status:PASS|WARN|FAIL|UNKNOWN,
  findings[], failed_rules[], provenance_refs[], created_at
}

DisagreementMetric {
  metric_id, outcome_space_id, implementation_ref,
  version, deterministic:true
}
```

Rules:

- LLM-002 gates MUST reference a versioned BenchmarkPolicy; calibration evidence must be traceable.
- VER-004 plausible outcomes MUST resolve to a concrete OutcomeSpace version.

- DomainPack validators MUST return ValidationReport, never an unstructured boolean/string.
- The disagreement metric MAY differ by domain, but MUST be declared, deterministic and versioned.

17.20 PriorArtSearchRecord

```
PriorArtSearchRecord {
  search_id, episode_id, intent, sources[], queries[],
  date_range?, retrieved_at, source_policy_version,
  result_count, deduped_work_count, limitations[], coverage_notes,
  external_source_record_ids[], inference_provenance_id?
}
```

17.21 ExternalSourceAdapter Interface Contract

所有外部 source provider 必須正規化成同一個 contract；SourceRouter 只依賴 capabilities / policy / normalized records，不知道 provider 實作細節。

```
ExternalSourceAdapter {
  provider_id() -> str
  capabilities() -> SourceCapabilities
  healthcheck() -> SourceHealth
  search(query: SourceQuery, policy: AccessPolicy) -> List[ExternalSourceRecord]
  fetch(locator: SourceLocator, policy: AccessPolicy) -> ExternalSourceRecord
  snapshot(locator_or_record, ref?, policy) -> ArtifactRef
}

ExternalSourceRecord {
  provider, source_type, canonical_locator, title?, authors/owner?,
  retrieved_at, visibility, version_ref?, content_hash?, snapshot_artifact_id?,
  rights_status?, license_metadata?, sensitivity, trust_class,
  metadata{}, raw_payload_ref?
}
```

17.22 GitHubConnector Contract

GitHubConnector 是 ExternalSourceAdapter 的一個實作，不是 core dependency。它負責 remote Git hosting knowledge retrieval；本地 Git commit provenance 仍由 LocalGitProvenance 處理。

```
GitHubConnector extends ExternalSourceAdapter {
  search_repositories(query, filters?)
  search_code(query, repo_scope?)
  fetch_repository_metadata(repo_locator)
  fetch_file(repo_locator, path, ref)
  fetch_commit(repo_locator, sha)
  fetch_release(repo_locator, tag_or_release_id)
  snapshot_repository(repo_locator, ref, paths?)
}

Required provenance:
  repository_identity
  visibility
  requested_ref
  resolved_commit_sha
  file_path (when applicable)
  retrieved_at
  snapshot/content hash
  license/rights metadata when detectable
  connector/auth scope identifier (never secret token)
```

Connector failure policy：rate limit、authentication failure、repository removed、ref drift、network unavailable 都必須回傳 structured connector error；不得由 LLM 猜測缺失內容。若一個 prior-art conclusion 依賴 GitHub material，且 material 已消失，狀態改為 SOURCE_UNAVAILABLE 並保留既有 snapshot（若合法且先前已保存）。

18. Final Repository Tree - 最終目錄樹架構

以下目錄樹以「長期 Laboratory Innovation Brain」為目標，同時避免一開始過度 microservice 化。core/ 為 domain-agnostic；domains/silicon_photonics/ 才包含 photonics physics 與 Lumerical specifics。

```

laboratory-innovation-brain/
├── README.md
├── pyproject.toml
├── uv.lock
├── .env.example
├── compose.yaml
├── Makefile
├── IMPLEMENTATION_STATUS.md
├── CHANGELOG.md
├──
├── docs/
│   ├── SAI_3.2.pdf
│   ├── architecture/
│   │   ├── epistemic_event_model.md
│   │   ├── execution_model.md
│   │   ├── source_policy.md
│   │   ├── security_boundary.md
│   │   └── domain_extension.md
│   ├── adr/
│   │   ├── ADR-0001-postgres-first.md
│   │   ├── ADR-0002-event-sourced-belief.md
│   │   ├── ADR-0003-claim-attestation-separation.md
│   │   ├── ADR-0004-intent-aware-source-policy.md
│   │   ├── ADR-0005-cost-vector.md
│   │   ├── ADR-0006-run-extractor-separation.md
│   │   └── ADR-0007-core-evidence-authority.md
│   ├── normative_statements.yaml
│   └── implementation/
│       ├── M0-foundation.md
│       ├── M1-epistemic-memory.md
│       ├── M2-silicon-photonics-tools.md
│       ├── M3-hypothesis-brain.md
│       └── M4-first-vertical.md
├──
├── configs/
│   ├── system.yaml
│   ├── storage.yaml
│   ├── model_routing.yaml
│   ├── privacy_policies.yaml
│   ├── source_policies.yaml
│   ├── evidence_authority.yaml
│   ├── budgets.yaml
│   └── domains/silicon_photonics.yaml
├──
├── src/lab_brain/
│   ├── bootstrap.py
│   ├── core/                                # domain-agnostic
│   │   ├── models/
│   │   │   ├── artifact.py
│   │   │   ├── source_work.py
│   │   │   ├── claim.py
│   │   │   ├── observation.py
│   │   │   ├── attestation.py
│   │   │   └── relation.py

```



```

├── retriever.py
├── inverted_retriever.py
├── source_router.py
├── retraction_erratum.py

├── cognition/                                # 6 core roles only
│   ├── supervisor.py
│   ├── evidence_researcher.py
│   ├── hypothesis_engine.py
│   ├── critic.py
│   ├── verification_planner.py
│   ├── novelty_auditor.py
│   ├── protocols/stake_adaptive_debate.py
│   ├── benchmark_policy.py
│   ├── model_router.py
│   ├── model_slots.py
│   └── inference_provenance.py

├── verification/
│   ├── planner.py
│   ├── sufficiency.py
│   ├── cost_vector.py
│   ├── capability_registry.py
│   ├── capability_router.py
│   ├── disagreement.py
│   ├── outcome_space.py
│   ├── validation.py
│   ├── authority.py
│   └── verification_gate.py

├── tools/
│   ├── registry.py
│   ├── contracts.py
│   ├── execution.py
│   └── job_adapter.py

├── domains/
│   ├── base.py
│   ├── registry.py
│   └── silicon_photonics/
│       ├── plugin.py
│       ├── condition_schema/
│       ├── authority_policy.py
│       ├── backend_validity.py
│       ├── validators/
│       ├── extractors/                # backend-agnostic Cj/Rs/Q/ER/...
│       ├── specialists/{semiconductor.py,optical.py,rf_system.py,thermal_control.py}
│       ├── capabilities/
│       ├── workflows/
│       ├── verification/
│       └── benchmarks/

├── tool_providers/lumerical/                # backend-bound run_* only
│   ├── bridge.py
│   ├── session.py
│   ├── license_pool.py
│   ├── charge.py
│   ├── mode.py
│   ├── fdtd.py
│   ├── interconnect.py
│   └── mock.py

├── integrations/
└── external_sources/

```

```

|   |   |— literature/
|   |   |— github/
|   |   |— patents/           # deferred after M4
|   |   |— web/              # deferred after M4
|   |   |— git/local_provenance.py
|   |
|   |— security/
|   |   |— actors.py
|   |   |— acl.py
|   |   |— classification.py
|   |   |— egress_guard.py
|   |   |— license_policy.py
|   |   |— audit_log.py
|   |
|   |— operations/
|   |   |— cost_ledger.py
|   |   |— budget_gate.py
|   |   |— review_queue.py
|   |   |— observability.py
|   |   |— backup_restore.py
|   |
|   |— interfaces/
|   |   |— cli.py
|   |   |— approvals.py
|   |
|— migrations/
|   |— 001_actors_projects.sql
|   |— 002_artifacts_sourceworks.sql
|   |— 003_claims_observations_attestations.sql
|   |— 004_relations.sql
|   |— 005_epistemic_events.sql
|   |— 006_jobs_runs.sql
|   |— 007_capabilities_costs.sql
|   |— 008_conditions.sql
|   |— 009_vectors.sql
|   |— 010_review_observability_benchmark_policies.sql
|   |
|— tests/
|   |— unit/
|   |— contract/
|   |— integration/
|   |— e2e/
|   |— security/
|   |— spec/test_requirement_traceability.py
|   |— spec/test_normative_statement_coverage.py
|   |— domains/silicon_photonics/
|   |
|— benchmarks/
|   |— core/{epistemic_replay,evidence_independence,source_policy,debate_diversity}/
|   |— silicon_photonics/{rs_contact_gap,mesh_artifact,condition_matching,hypothesis_discrimination}/
|   |
|— fixtures/
|— scripts/{bootstrap_dev.py,migrate.py,ingest.py,replay_beliefs.py,run_benchmark.py,verify_environment.py}
|— deployment/local/           # lab_server / air_gapped deferred until needed

```

18.1 些資料夾第一版先不實作

- learning/sft、preference、offline_rl：先建立 trajectory dataset 再做。
- interfaces/dashboard：CLI / notebook 先跑通 verification loop，UI 後做。
- integrations/lims：若實驗室沒有 LIMS，不要為了完整性先做。

- domains/silicon_photonics/fabrication、thermal_control：先留 interface / schema，Ring v1 可延後。
- retrieval/graph_paths：Phase 1 用 relation tables 即可，無需專用 graph engine。

19. GitHub Attribution Matrix - 每一模組從裡取材

目標系統模組	參考來源	取什麼	不取什麼	目標系統新設計
Hypothesis Generation/Reflection/Evolution	Kaimen Co-Scientist / Google AI Co-Scientist pattern	Generation, Reflection, Ranking, Evolution, Proximity	Elo 決定 truth	用 tournament 分配 simulation attention ; truth 交給 evidence
Scientific Evidence Retrieval	FutureHouse PaperQA2	scientific search + evidence gathering + citation grounding	把 QA output 當 memory	增加 condition normalization 、 evidence object 、 provenance
Cross-domain Graph Reasoning	MIT SciAgents	graph paths / multi-agent scientific connection	大型 ontology upfront	Phase 2 再引入 graph reasoning ; v1 relation tables
Photonic Execution Loop	Flexcompute AutoPhotonicDesign	DRC → simulate → analyze → journal	單 scalar FoM keep/discard	改成 hypothesis-discriminating experiment + multi-fidelity
Research Tree Exploration	Sakana AI Scientist v2	best-first / progressive branching	無限制 tree expansion	budget-aware hypothesis portfolio ; 昂貴 solver cost model
Research Team Protocol	Virtual Lab	PI / specialist meeting concept	自由聊天	獨立 position + cross-examination + experiment proposal
Workflow Separation	Agent Laboratory	literature / experiment / report stage separation	paper writing 當核心	用 evidence/hypothesis/run stage isolation 防 confirmation bias
Perspective-Guided Questions	STORM / Co-STORM	多視角問題與 shared conceptual space	把 mind map 當 scientific truth	用於 research gap / question discovery
Experimental Feedback	FutureHouse Robin	實驗資料回流 candidate generation	生醫-specific pipeline	抽象成 observation→belief update
Principle-Guided Discovery	PrIM	physics/chemistry principles + virtual experiments	property optimizer = discovery	把 principles 轉 validators/falsifiers
Research Infrastructure	ARI	MCP plugins 、 claim-evidence gate 、 reproducibility/BFTS pattern	完整 paper automation	借 tool plugin/provenance/evaluation or pattern
Lumerical Bridge	lumerical-mcp	lumapi multi-product session pattern	arbitrary eval_script	typed domain tools + immutable manifests
Document Parsing	Docling	layout-aware scientific ingestion	parser-specific data model	adapter → Laboratory Brain normalized artifact/evidence
Stateful Orchestration	LangGraph	explicit state graph / checkpoint idea	raw data in graph state	只傳 entity IDs / structured summaries
External Source	GitHub remote	repository/code	不讓 cognition	ExternalSource

目標系統模組	參考來源	取什麼	不取什麼	目標系統新設計
Connectivity / GitHub	repository interface pattern + provider-agnostic adapter architecture	discovery、ref-pinned retrieval、commit provenance、capability-based connector registry	直接依賴 GitHub SDK；不把 README/stars 當 scientific truth；不假設可複製任何程式碼	Adapter + GitHubConnect or + SourceRouter + privacy/rights/provenance gate，可替換成 GitLab/Gitea/其他 provider

20. Risk Register - P0/P1/P2 風險與防線

Priority	風險	後果	防線
P0	Epistemic contamination	LLM inference 被寫成 evidence , 數月後自我引用成「已知事實」	schema-level epistemic type + admission gate + provenance audit
P0	Private data leakage	NDA PDK / unpublished idea 被送外部 search/model	local privacy gateway + labels + private mode + egress logs
P0	Evaluator corruption	Agent 改 metric extractor / solver config 讓自己「看起來變好」	typed tools + immutable evaluator + hidden validation gate
P0	Unreproducible runs	simulation 沒記 version/mesh/material/script commit	mandatory run manifest + artifact hash + replay tests
P1	Overengineering	過早部署多 DB / graph / swarm , 團隊無法維護	PostgreSQL-first + measured scale-out + phase gates
P1	Groupthink	多 Agent 互相附和造成假共識	stake-adaptive independent positions + Critic inverted retrieval + bundle-divergence/diversity metrics + external verification
P1	Condition mismatch	不同 wavelength/bias/platform paper 被錯誤合併	condition-aware evidence / structured filters
P1	Simulator overfit	Agent 找 solver artifact 而非 robust design	multi-fidelity / hidden corners / independent rerun / measurement
P1	Memory write friction	研究員被迫手填大量 schema , 最後停止使用	non-invasive ingestion + auto extraction + only decision/root-cause annotation
P1	Historical bias	長期 memory 讓系統只重複實驗室既有思路	external novelty audit + exploration budget + graph cross-domain search
P2	Model/provider churn	API / model names 迅速變動	model abstraction + capability routing, not model-name coupling
P2	Storage product churn	S3/vector/graph product 維護狀態改變	protocol/interface abstraction + exportable open formats
P2	Copyright/rights	外部 full text cache policy 不一致	rights metadata + source-specific retention; no blanket fair-use assumption
P0	Evidence extraction	抽取器為填滿 schema	field-level status +

Priority	風險	後果	防線
	hallucinated completion	自行補 doping/bias/geometry, 污染後續 condition-aware reasoning	UNKNOWN/NOT_REPORTED + targeted enrichment + human escalation
P0	Low-fidelity false falsification	coarse mesh / simplified model 的 numerical artifact 被當成物理反例, 正確 hypothesis 被過早淘汰	fidelity-aware epistemic authority + PROVISIONAL_CHALLENGE + standard/validation fidelity escalation
P1	Simulator treated as ground truth	模型/材料/mesh 系統誤差被忽略, sim-to-real conflict 被錯誤抹平	EvidenceAuthority + AuthorityPolicy.compare + backend validity + conflict preservation; no fixed simulator>measurement ordering
P1	Tacit-knowledge capture friction	教授/學生不願填複雜表單, Heuristic layer 長期空洞	authorized-source Heuristic Miner + candidate certificate + approve/correct/reject workflow
P1	Graph backend coupling	Agent/service 直接依賴 Raw SQL CTE, 未來 traversal workload 變化難以遷移	GraphRepository abstraction + conformance/regression tests + benchmark-triggered migration
P1	External source/provider	Agent code 直接綁 GitHub/某 paper	ExternalSourceAdapter + connector

Priority	風險	後果	防線
	coupling	API , provider 改版或換平台時 cognition 層一起壞	registry + contract tests
P1	Remote source drift / deletion	GitHub branch 、 README 、 release 更新後無法重現當時 reasoning	ref/commit pinning + snapshot hash + SOURCE_UNAVAILABLE lifecycle
P0	Private repository / query leakage	授權 private repo 或 unpublished concept 被混入不受控外部 search	explicit auth scope + allowlist + egress policy + audit; no secret in provenance
P0	Belief revision not replayable	錯誤 evidence/extractor 被發現後無法重新推導目前 belief ; audit/trajectory dataset 失效	append-only BeliefRevisionEvent + projector + quarantine/replay tests
P0	Evidence independence inflation	preprint/正式版/ review/轉引被重複算成獨立支持	Claim/SourceWork identity + Attestation + SAME_WORK_AS/CITES + independence policy
P0	LLM reasoning without provenance	Hypothesis/ critique/support judgment 無法重現	InferenceProvenance mandatory on every LLM-produced scientific object/relation
P0	No actor/ACL enforcement	NDA 、 approval 、 egress 無法回答「誰有權」	Actor + ProjectMembership + sensitivity clearance + default-deny labeling
P0	Cost / license	多角色/長時	CostLedger +

Priority	風險	後果	防線
	runaway	solver/API 造成不可控 費用與資源爭用	BudgetGate + Capability estimate_cost + license pool/job queue
P0	Async scientific workflow missing	FDTD/ measurement/ fabrication 使同步 episode 永久阻塞	Job entity + AWAITING_* states + suspend/resume events
P1	JSONB condition semantic drift	多年後 condition- aware retrieval 靜默 退化	versioned Domain ConditionSchema + DB validation + migration/backfill policy
P1	Embedding-space mixing	換 embedding model 後 cosine 距離失真	model-version filtered retrieval + dual-index re-embed migration
P1	Source corroboration / retraction blind spot	撤稿或同源重複資料持 續支持 belief	source-work dedup + retraction/erratum check + PriorArtSearchRecord
P1	Human review backlog	所有 uncertainty 都流 向教授造成永久 PENDING	ReviewItem schema + ReviewQueue capacity/SLA/expiry + human-review Capability availability fed into Planner
P0	Unruled belief transition	Event 可重播但狀態仍 由 LLM 任意決定	versioned TransitionPolicy + deterministic evaluate + EPI-005

Priority	風險	後果	防線
			test
P0	Authority comparison undefined	重大 contradiction gate 無法決定	AuthorityPolicy partial-order comparator + INCOMPARABLE handling
P1	Spec prose not enforced	散文 MUST 沒 requirement/test , Coding Agent 不實作	Normative Statement Registry + T-SPEC-002 coverage lint

21. Day-to-Day Lab Operation - 完成後每天怎麼運作

以下情境描述 Laboratory Innovation Brain 完成後在實驗室的日常角色。重點不是「每件事都自動」，而是所有重要研究事件都回到共同 scientific memory，讓下一個人能繼續。

21.1 早上：新論文進來

1. External connector 發現或研究員加入新 paper。
2. Parser 保存 artifact、DOI、版本與 rights metadata；抽取 section/table/figure/equation。
3. Evidence Normalizer 產生 claim + condition + locator，並與既有 hypotheses / claims 做 condition-aware match。
4. 若 paper contradicts active hypothesis，Epistemic State 必須建立 ReviewItem(subject_type=CONFLICT)，保留衝突與來源，不得直接覆蓋舊結論。
5. Evidence Agent 在下次相關 episode 會優先提示這個新衝突。

21.2 下午：學生完成一組 CHARGE sweep

1. Typed tool 自動產生 Run manifest：project hash、bias、AC frequency、mesh、solver version、script commit。
2. Raw output / plots / arrays 進 Artifact Layer；metric extractor 產生 $C_j(V)$ 、 $R_s(V)$ Evidence。
3. 系統比對原 hypothesis predictions。
4. 若 R_s trend 異常，Case-Based Retriever 找歷史相似 failure；Critic 產生 contact / mesh / extraction 等 competing causes。
5. Verification Planner 建議最便宜的區辨測試；學生或 Supervisor 核准後執行。
6. 確認 root cause 後，FailureAnalysis 與 Heuristic 更新；未來再遇到同症狀會先查這些。

21.3 Meeting：教授提出「能不能把熱漂移與控制一起做？」

1. Supervisor 建立 ResearchEpisode 與 privacy scope。
2. Evidence Function 分 semiconductor / optical / thermal / control perspectives 搜集內外部證據。
3. 各 specialist 獨立提出 position，不先互相看答案。
4. Hypothesis Engine 形成 competing mechanisms / design hypotheses。
5. Critic 要求每個 hypothesis 有 falsifier 與 confounder。
6. Verification Planner 選擇能以 MODE / INTERCONNECT 最快區分的 simulation。

7. 結果回寫後，meeting 紀錄只需補充人類決策理由與下一步。

21.4 一年後：新學生加入

新學生不需要從零翻學長資料夾。他可以問：「我們過去所有 ring modulator Rs 異常有哪些原因？哪幾次是 contact gap、哪幾次是 mesh？各自怎麼驗證？」Laboratory Innovation Brain 應回傳 evidence-linked episodes，而不是一段沒有來源的總結。這才是 Laboratory Innovation Brain 的傳承價值。

22. Acceptance Criteria - Phase-1 System Acceptance

層級	Acceptance Criteria
Data/Provenance	任一引用的 internal simulation metric 可在 2-3 hops 內追到 Run + Artifact hash + solver settings。
Evidence	系統不允許 INFERRED / HYPOTHESIS 物件被當成 measured/simulated evidence。
Retrieval	對 ring modulator benchmark，可依 wavelength/bias/device conditions 過濾不相容 evidence。
Hypothesis	每個 admitted hypothesis 具有 mechanism、prediction、falsifier、confounders、minimum test。
Verification	Planner 只選 declared plausible outcome 可改變 policy decision 的 sufficient actions；Pareto 後使用 versioned deterministic SelectionPolicy。
Simulation	至少 CHARGE + MODE + INTERCONNECT 具有 typed tools 與 reproducible run manifest。
Memory	同一真實 failure case 再出現時，系統可找回歷史 episode 並減少無效 solver calls。
Privacy	Private Mode 的受限資料不產生外部 network request；Audit 可驗證。
Benchmark	能比較 Single LLM / RAG / Structured Evidence / Full Laboratory Innovation Brain 的 root-cause accuracy、solver cost、unsupported claim rate。
Human Value	教授/研究生能從 episode lineage 快速理解「做過什麼、為什麼、結果、下一步」，不依賴原作者在場。

層級	Acceptance Criteria
Extraction Integrity	關鍵 evidence 欄位無法確認時可輸出 UNKNOWN/NOT_REPORTED；測試集中不得以模型常識偷偷補值。
Fidelity Integrity	Exploratory/low-fidelity evidence 不可單獨將核心 Hypothesis 標為 REJECTED；需能觸發 escalation。
Evidence Authority	任一重大 belief revision 所用 evidence 可由 AuthorityPolicy 比較；INCOMPARABLE 不得硬排序。
Tacit Knowledge Governance	Heuristic Miner 只從 approved sources 產生 Candidate Heuristic；未經人類核准不得升級為 lab rule。
Graph Abstraction	Cognition/retrieval 不得直接依賴 backend-specific SQL/Cypher；GraphRepository contract 必須通過 conformance/regression tests。
External Sources	Literature/GitHub/Web/ELN 等來源透過同一 ExternalSourceAdapter boundary；停用 GitHub connector 後 core cognition 仍可啟動。
GitHub Provenance	任何被引用的 GitHub file/code/release 能追到 repository identity + resolved commit/ref + retrieved_at + content/snapshot hash。
Private GitHub Security	未配置明確 authentication + allowlist/project scope 時，系統不得讀 private repo；restricted context 不得被用於 uncontrolled public search。
Epistemic Replay	隔離一筆曾導致 belief transition 的 Attestation 後，可 replay events / relations 並得到可驗證的新 EpistemicState；不得靠手改 current status。
Claim Independence	同一 work 的 preprint + journal + review fixture 不得被計成三個獨立 corroborations。

層級	Acceptance Criteria
Inference Provenance	任一 Hypothesis/Critique/SUPPORTS judgment 可追到 model/version + prompt version + evidence bundle hash + source policy。
Actor/ACL	未授權 actor 無法讀取/egress restricted project；所有 approval 與 egress 有 actor_id audit。
Async Jobs	mock 6-hour job 可 submit→AWAITING_JOB→event→resume；duplicate callback 不產生重複 run (idempotent)。
Cost Governance	LLM/tool call 前 budget gate 可預估/檢查成本；超限 fail-closed 或 human approval。
Condition Versioning	每筆 condition-aware record 具 schema version；不符合 schema 的 write 被拒絕或 quarantine。
Sim-to-Real Extractor	同一 backend-agnostic metric extractor 可接受 simulation array 與 measurement array fixture，輸出可比較且記 normalization/method。
Source Intent Policy	DIAGNOSIS 與 NOVELTY_AUDIT 使用不同 SourcePolicy；Critic 可產 inverted evidence bundle。
Spec Integrity	Requirement ID 唯一，且每個 normative requirement 至少映射一個 test；不存在 test 指向未知 ID。
Transition Governance	所有 scientific state transition 由 TransitionPolicy 允許；LLM 不可直接改 Hypothesis status。
Bundle Reproducibility	任一 LLM scientific output 可追到 canonical EvidenceBundle hash，重建相同 ordered evidence set。
Human Capacity	human-review Capability availability 會隨

層級	Acceptance Criteria
	ReviewQueue capacity/depth 改變，Planner 不把人類審核視為免費無限資源。
Operational Guardrails	secret/license/embedding-version/ retraction/observability 至少各有 Requirement ID + automated/fixture test。

23. Agent Implementation Protocol - 讓 Coding Agent 直接照規格施工

本章是 SAI 3.2 與前版最大的差異：它把 architecture vision 轉成 Agent 可以執行、Human 可以逐步驗收的施工契約。Agent 不應一次「把整個系統做完」；應按 milestone 建立可運行 vertical slices，每一個 requirement 都有輸出檔案、測試與狀態。

23.1 Implementation Agent Operating Rules

ID	Normative requirement	Definition of Done
AGT-001	Agent MUST 先讀 SAI 3.2、IMPLEMENTATION_STATU S.md、現有 ADR 與目前 milestone，再修改程式。	每次工作輸出列出讀過的規格版本與 requirement IDs。
AGT-002	Agent MUST 一次完成一個可驗收 slice；不得同時大改 storage、schema、agents、UI。	PR/patch 可被單獨測試與回退。
AGT-003	Agent MUST 更新 IMPLEMENTATION_STATU S.md：完成、部分完成、blocked、測試結果、下一步。	文件與實作狀態一致。
AGT-004	Agent MUST NOT silent schema drift。任何 migration 都要有 schema version 與向後/向前策略。	migrations + contract tests 通過。
AGT-005	若偏離 SHOULD 或變更 MUST，Agent MUST 建 ADR；碰到 P0 contract 必須等待 Human approval。	ADR 被記錄並列出 affected requirements。
AGT-006	Agent SHOULD 優先 CLI + tests，不應先做 Dashboard。	無 GUI 仍可跑核心 vertical slice。
AGT-007	Agent MUST 提供 mock/fake boundaries，使 CI 在沒有 Lumerical license、外網或雲	CI 可以用 mock simulator 跑核心 contract/e2e tests。

ID	Normative requirement	Definition of Done
	端 LLM 時仍可驗證核心邏輯。	
AGT-008	Agent MUST implement external provider access behind ExternalSourceAdapter; cognition/domain code MUST NOT call GitHub/Web/Literature provider SDK directly.	Connector contract tests pass; disabling/replacing one provider does not require cognition source changes.
AGT-009	Agent MUST NOT mutate EpistemicState directly; all scientific state transitions go through BeliefRevisionEvent/event projector.	Static/contract tests reject direct repository update path.
AGT-010	Agent MUST attach InferenceProvenance to every LLM-produced hypothesis/critique/relation judgment.	Missing provenance prevents admission/transition.
AGT-011	Agent MUST keep core domain-agnostic: simulator-specific validity fields belong to DomainPack/backend schemas.	Core model imports contain no mesh/CHARGE/MODE-specific fields.
AGT-012	Agent MUST implement long-running external/tool actions as Jobs with idempotent resume events.	Mock delayed job e2e passes and can recover after process restart.
AGT-013	Agent MUST enforce budget/ACL before external/model/tool execution, not after.	Security/cost tests show no side effect when gate denies.
AGT-014	Agent MUST separate	Same extractor contract

ID	Normative requirement	Definition of Done
	backend-bound run_* from backend-agnostic extract_* metric logic.	passes simulation and measurement fixtures.
AGT-015	Agent MUST NOT resolve spec conflicts via “newer-looking” prose; conflicting normative contracts block the slice and require spec issue/ADR.	Spec lint / review must report direct conflicts before implementation.
AGT-016	Agent MUST use TransitionPolicy/AuthorityPolicy operators; LLM output alone cannot cause scientific status transition.	Contract tests reject direct status assignment from cognition path.
AGT-017	Agent MUST construct canonical EvidenceBundle before any scientific LLM call and persist bundle hash in InferenceProvenance.	LLM call wrapper refuses missing bundle_id/hash.

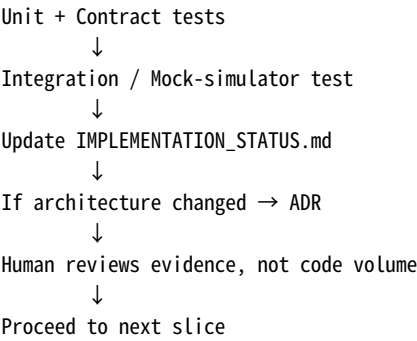
23.2 Requirement ID Namespace

SYS-xxx system/core architecture
 ART-xxx artifact/source-work/provenance
 EVI-xxx claim/evidence/retrieval/source routing
 EPI-xxx belief events / epistemic projection / hypothesis
 VER-xxx verification planning / capability / authority
 SIM-xxx simulator/backend contracts
 DOM-SP-xxx silicon photonics domain
 SEC-xxx actor/ACL/privacy/egress
 OPS-xxx job queue / observability / review operations
 COST-xxx cost ledger / budget gates
 SRC-xxx external source / intent-aware routing
 LLM-xxx model router / inference provenance
 EXT-xxx domain extension
 TST-xxx meta/spec/CI requirements

23.3 Mandatory Development Loop

```

Read spec + current status
↓
Select next unblocked Requirement IDs
↓
Implement smallest coherent slice
↓
  
```



23.4 Human Steering Points

Human 不需要逐行指揮 Agent 寫 code；Human 應控制 scientific semantics 與 scope。下列決策不可默默交給 Agent 自行定義：Evidence 何時可升級成 verified、什麼 fidelity 足以反駁 hypothesis、哪些 external connector 可接、哪些資料屬於 NDA/CONFIDENTIAL、Silicon Photonics benchmark ground truth、以及何時將某個 experimental heuristic 升級成 hard validator。

Human 決策點	Agent 可自行做	Agent 不可自行做
Database implementation	index、query optimization、repository code	把 Evidence semantics 改成單純 vector chunk；移除 provenance。
Lumerical adapter	session management、serialization、retry	改變物理模型或 validator 標準來讓測試過。
Agent prompts/models	prompt wording、model adapter	用 LLM vote 取代 simulator/measurement evidence adjudication。
Domain extension	建立符合 DomainPack contract 的新 plugin	在 core 寫 domain-specific if/else。
Security	policy engine implementation	自動把 restricted context 送到外部 LLM/search。

23.5 Normative Architecture Coverage Rule

§ 6- § 16 為 normative architecture。為維持人類可讀性，不要求每一句 MUST 都在正文尾端塞 ID；改用 Normative Statement Registry 讓每條硬性規範可唯一映射到 Requirement ID、Test ID 或明確 DEFERRED rationale。T-SPEC-001 檢查 ID/traceability；T-SPEC-002 檢查 normative-statement coverage。未登錄的 hard MUST 視為 spec defect，不得被 Coding Agent 默默忽略。

23.6 Normative Statement Registry

Registry 可由文件旁的 machine-readable YAML/JSON 維護，至少含 statement_key、section、normative_level、requirement_id、test_ids、deferred_rationale。這比在每個人類可讀句子後硬塞 Requirement ID 更穩定，也能讓 CI 驗證「散文桶」不再出現。

```
normative_statements:
  - key: belief.transition.policy
    section: 8.2.1
    level: MUST
    requirement_id: EPI-005
    tests: [T-EPI-005]
  - key: authority.incomparable.review
    section: 8.2.1
    level: MUST
    requirement_id: EPI-004
    tests: [T-EPI-004]
  - key: critic.inverted_retrieval.high_stakes
    section: 7.2
    level: MUST
    requirement_id: SRC-002
    tests: [T-SRC-002]
  - key: review.capacity.planner
    section: 14.4.1
    level: MUST
    requirement_id: OPS-002
    tests: [T-OPS-002]
  - key: benchmark.policy.calibration
    section: 15.4
    level: MUST
    requirement_id: LLM-002
    tests: [T-LLM-002]
```

24. Domain Extensibility Contract - 先光子、核心不寫死

SAI 3.2 的實作優先級不是「先做一個抽象到什麼都能支援的框架」，而是「先把 Silicon Photonics 做深，但把 dependency direction 與 extension points 設對」。換句話說：先 vertical-first，再 interface-stable；不要先做空泛 universal ontology。

24.1 Core vs Domain Boundary

Core MUST know	Core MUST NOT know	Silicon Photonics DomainPack owns
Artifact / SourceWork / Claim / Observation / Attestation / Relation / Hypothesis / Episode / Job / Actor / Capability / CostVector / BeliefRevisionEvent	Cj、Rs、Q、FSR、ring radius、PN doping、 CHARGE/MODE command names、mesh convergence field names	光子 condition schema、 EvidenceAuthority policy、 backend validity、physical validators、metric extractors、specialist roles、Lumerical bindings、 benchmarks
InferenceProvenance、 SourcePolicy contract、 EvidenceAuthority abstraction、ACL、budget、 event replay	「measurement 永遠最高」、 「1550 nm 是預設」、任何 solver-specific hierarchy	波長/偏壓/溫度/幾何/doping/ driver-load semantics、 mesh/solver/calibration validity rules
Tool registry、 VerificationAction types、 DomainPack registry、 graph/evidence repositories、cost/fidelity concepts	某一個 solver 的參數細節；不 得假設所有 domain 都有 simulator	CHARGE/MODE/FDTD/ INTERCONNECT typed adapters、domain analytical checks、measurement bindings

24.2 Dependency Rule

ALLOWED:

domains.silicon_photonics → core
domains.silicon_photonics → tools/simulators interfaces
cognition → DomainPack registry/interface

FORBIDDEN:

core → domains.silicon_photonics
evidence generic layer → ring-specific constants
orchestration → direct CHARGE/MODE imports
database schema → hard-coded ring-only columns unless isolated in domain extension tables

24.3 DomainPack Protocol

```
class DomainPack(Protocol):
    id: str
    version: str

    def register_condition_schema(self, registry): ...
    def register_condition_comparator(self, registry): ...
    def register_evidence_authority_policy(self, registry): ...
    def compare_authority(self, a, b) -> AuthorityComparison: ...
    def register_backend_validity_schemas(self, registry): ...
    def register_validators(self, registry): ...
    def register_metric_extractors(self, registry): ...
    def register_capabilities(self, registry): ...
    def register_tools(self, registry): ...
    def register_specialists(self, registry): ...
    def register_workflows(self, registry): ...
    def register_benchmarks(self, registry): ...
    def register_disagreement_metrics(self, registry): ...

    def validate_attestation(self, attestation) -> ValidationReport: ...
    def compare_conditions(self, a, b) -> ConditionMatch: ...
    def declared_outcome_space(self, action, hypothesis) -> OutcomeSpace: ...
    def disagreement_metric(self, outcome_space) -> DisagreementMetric: ...
```

24.4 Deferred Extension Examples (Informative, not M0-M4 scope)

以下領域只作為 extension-boundary examples，不是 v3.2 M0-M4 交付範圍，也不是固定優先順序。真正擴充時以研究需求、共享 evidence/tool boundary 與 EXT-001 benchmark 決定。

Priority	Extension Domain	Why it is adjacent	Core reuse
1	Optical Communication / WDM System	ring/MZM/PD/link budget/BER 與現有 INTERCONNECT、system metrics 直接相連	Evidence/ Hypothesis/ Episode/ VerificationPlan 幾乎全部重用
2	RF / High-Speed Electronics	modulator electrode、impedance、S-parameter、driver/load 是高速 SiPho 的直接瓶頸	Numerical store、condition matching、active verification planning
3	Thermal / Control	ring resonance drift、heater locking、temperature perturbation	multi-fidelity + causal hypothesis + control verification actions
4	Semiconductor	PN junction、	CHARGE workflow、

Priority	Extension Domain	Why it is adjacent	Core reuse
	Device / Process	contact、implant、mobility、process corners	failure diagnosis、fabrication evidence
5	Photonic Packaging / Test	coupling loss、alignment、fiber attach、measurement calibration	measurement provenance、failure analysis、sim-to-real
6	Materials / Heterogeneous Integration	SiN/III-V/LN/Ge 等新材料系統	核心保留，但需要新的 condition/entity/validator/tool packs

24.5 Extensibility Acceptance Test

EXT-001：在不修改 core models、core repositories 與 orchestration state machine 的前提下，加入一個 ToyDomain 或第二個真實 domain，並能註冊 condition schema、validator、tool 與 benchmark。若必須修改 core 才能加入第二 domain，代表 extension boundary 失敗。

25. Silicon Photonics First Vertical - 第一條可驗收端到端實作

第一版不要從「自動讀全世界論文 + 多 Agent + 完整 Lumerical 全家桶」開始。第一條 vertical slice 應以已知可驗證案例串起 Memory、Evidence、Hypothesis、Verification Planner、Typed Tool、Verification Gate 與 Failure Analysis。重點不是一定要跑最多 solver，而是系統能否先用便宜檢查縮小原因，再只跑必要的 targeted verification。推薦第一案例：PN junction Rs 異常診斷。

25.1 Vertical Slice VS-SP-001 - Rs Bias-Insensitive Anomaly

INPUT
Existing CHARGE project / normalized mock equivalent
Bias sweep: 0 → reverse bias
Observation: Cj trend plausible; Rs extremely high and weakly bias-dependent

EXPECTED RESEARCH LOOP
Artifact ingest
→ Run Manifest
→ Evidence extraction (Cj(V), Rs(V), solver validity)
→ Anomaly detection
→ competing hypotheses
 H1 access/contact discontinuity
 H2 mesh / convergence artifact
 H3 contact/material/normalization model issue
→ experiment candidates
 connectivity validator / mesh sensitivity / model check
→ choose cheapest discriminative check
→ run typed tool
→ confirmed finding
→ FailureAnalysis + Heuristic candidate
→ belief update + provenance graph

25.2 Required Typed Tools for First Vertical

Tool ID	Typed operation	Minimum output
DOM-SP-TOOL-001	inspect_device_connectivity	regions, contacts, connectivity violations, artifact refs
DOM-SP-TOOL-002	run_charge_bias_sweep	bias points, run manifest, convergence status, raw artifact refs
DOM-SP-TOOL-003	extract_cj_rs	Cj_per_length, Rs_per_length, units, method, frequency/bias conditions
DOM-SP-TOOL-004	run_mesh_sensitivity	mesh variants, extracted

Tool ID	Typed operation	Minimum output
		metrics, stability delta, validity report
DOM-SP-TOOL-005	validate_expected_trends	trend checks with domain rule version; never changes raw evidence

25.3 First Vertical Requirements

Requirement	MUST
ART-001	每個輸入 project/script/result 必須有 immutable artifact ID + hash + source + timestamp。
SIM-001	每次 CHARGE run 必須保存 solver version、project hash、mesh/bias/config、exit/convergence status。
EVI-001	Cj/Rs evidence 必須含 unit、normalization basis、bias/frequency conditions 與 extraction method。
EVI-002	缺失欄位必須是 UNKNOWN/NOT_REPORTED，不得由 LLM 補常見值。
EPI-001	至少維護 2 個 competing hypotheses；單一看似合理原因不得直接被升級成 confirmed root cause。
VER-001	下一個 verification action 必須有 predicted discriminatory outcome + estimated cost；若較便宜 evidence 已足夠，不得無理由升級到 simulator。
SIM-002	coarse/low-fidelity contradiction 只能標記 CHALLENGED；要 REJECT hypothesis 必須過 standard/validation fidelity gate。
EPI-002	confirmed root cause 必須可以 trace 回 Run/Evidence/Artifact；LLM statement 不可

Requirement	MUST
	作為證據。
DOM-SP-001	Silicon photonics rule/validator 版本必須記錄；core 不得內建 Rs 趨勢規則。
TST-001	同一案例要能以 mock dataset 在 CI 重播，並以真實 Lumerical project 在有 license 的環境重播。
SRC-001	所有外部知識取得經 ExternalSourceAdapter/SourceRouter；provider-specific SDK 不得滲入 cognition/domain core。
GH-001	GitHubConnector 至少支援 public repo discovery + ref/commit-pinned file fetch + normalized provenance record。
GH-002	private repo access 需 explicit auth/allowlist/security policy；未授權時必須 fail closed。
GH-003	GitHub technical/prior-art evidence 不得自動升級為 peer-reviewed scientific evidence；repo/ref/commit 必須保存。
EPI-003	所有 hypothesis status 變更必須產生 BeliefRevisionEvent；EpistemicState 可由 event replay 重建。
EVI-004	同一 underlying work / derived source 必須經 Claim/SourceWork resolution；corroboration 不得重複計數。 DEPENDENCE_UNKNOWN contributes 0 to min_independent_attestations until resolved.
LLM-001	所有 LLM scientific outputs 必須帶 InferenceProvenance：model/version、slot、prompt version、evidence bundle hash。
SEC-002	所有 actor/approval/egress/private-source

Requirement	MUST
	access 必須經 Actor+ACL；新 internal artifact sensitivity 預設 fail-closed。
OPS-001	long-running tool/external actions 必須 Job 化並支援 suspend/resume/idempotent callback。
COST-001	每次 LLM/tool action 前必須經 BudgetGate；CostLedger 至少記 wall-clock/tokens/money/license-seat estimates。
VER-002	Verification Planner 必須使用 Capability descriptors；sufficient candidate 有明確 state-transition/blocked-conflict criterion。
VER-003	Verification cost 使用 CostVector；不可只存 normalized_cost 作唯一決策依據。
DOM-SP-002	Cj/Rs/Q/ER 等 extractor 必須 backend-agnostic；simulation 與 measurement 共用定義與 normalization provenance。
EVI-005	Domain condition records 必須含 conditions_schema_version；ConditionMatch 語意由 DomainPack 註冊。
SRC-002	SourcePolicy 依 research intent 切換；當 decision.stakes >= policy threshold，Critic MUST 執行 inverted retrieval、保存 inverted EvidenceBundle，否則該 decision 不得進入 BELIEF_REVISION。
TST-002	規格一致性測試必須檢查 requirement ID 唯一與 requirement↔test traceability。
EPI-004	DomainPack MUST expose AuthorityPolicy.compare returning STRONGER/WEAKER/EQUIVALENT/INCOMPARABLE. INCOMPARABLE at a required transition gate MUST yield NEED_HUMAN_REVIEW + ReviewItem(AUTHORITY_CONFLICT); no

Requirement	MUST
	silent promotion/rejection.
EPI-005	Hypothesis state transition MUST be authorized by versioned TransitionPolicy and emitted as BeliefRevisionEvent; LLM cannot directly mutate status.
EVI-006	Every scientific LLM call MUST use a canonical EvidenceBundle with deterministic hash over ordered attestation IDs + query/policy/condition snapshot.
VER-004	Plausible outcomes MUST come from a declared, versioned OutcomeSpace and pass explicit constraints/ValidationReport checks; planner may not invent outcomes.
VER-005	After Pareto filtering, planner MUST apply a versioned deterministic SelectionPolicy; identical input+policy returns identical ranked plan.
SEC-003	Immutable ingestion MUST run secret scan first; suspected credentials go to quarantine/redaction flow, not permanent normal artifact storage.
SEC-004	External code MUST carry license_class; UNKNOWN/COPYLEFT source code is blocked from code-generation context unless project policy explicitly permits.
EVI-007	Vector retrieval MUST filter by compatible embedding model/version; embedding migration uses dual-index + verified cutover.
EVI-008	Major belief revision from external reported evidence SHOULD pass source-work version/retraction/erratum check; check status MUST be recorded.
OPS-002	Human ReviewQueue MUST have capacity,

Requirement	MUST
	stakes, SLA/expiry and feed availability/earliest_available_at into human-review Capability.
OPS-003	Execution observability MUST persist trace/span contract linking episode -> retrieval/LLM/job/run/artifact with status and cost refs.
TST-003	Spec CI MUST verify normative-statement registry coverage: every registered MUST/SHOULD maps to a unique Requirement ID and at least one test or explicit DEFERRED rationale.
SYS-001	Core scientific state path MUST be Artifact/SourceWork -> Claim/Observation -> Attestation -> RelationJudgment -> TransitionPolicy -> BeliefRevisionEvent -> EpistemicStateProjection.
LLM-002	Structured Debate MUST collect diversity/bundle-divergence/cost metrics; hard gates use a versioned BenchmarkPolicy calibrated on a fixed domain benchmark before enforcement.
EVI-003	LLM-generated interpretation/inference MUST NOT be admitted as OBSERVED/SIMULATED/MEASURED evidence; it remains an inference/RelationJudgment with InferenceProvenance.
SEC-001	RESTRICTED_NDA/CONFIDENTIAL context MUST NOT leave the approved boundary; external connector/model egress requires policy + Actor clearance, otherwise fail closed and emit audit evidence.
EXT-001	A ToyDomain or second real DomainPack

Requirement	MUST
	MUST be addable through plugin registration without modifying core models, core repositories, or orchestration state machine.

25.4 Vertical Slice Definition of Done

VS-SP-001 只有在以下條件全部成立才算完成：CLI 可從一個 project/fixture 建立 episode；raw artifacts 被 hash；Cj/Rs 被抽成 structured evidence；系統提出並保存 competing hypotheses；Verification Planner 先檢查 historical case / connectivity / extraction consistency 等低成本動作，再在需要時升級 simulator；mock/real backend 都走 typed contract；結果觸發 fidelity-aware update；confirmed root cause 可追溯到 Run/Evidence/Artifact；同類失敗形成可審核 heuristic candidate。

25.5 Second Vertical - Ring Modulator Mechanism Chain

完成 VS-SP-001 後，第二條 vertical 才擴充到 CHARGE → MODE → INTERCONNECT：以 bias-dependent Cj/Rs、neff/loss、resonance/ER/IL/BER/eye 為 observation chain，測試 Laboratory Brain 是否能區分「電性 RC 改善」「光學 overlap/loss 改變」「操作 detuning」「thermal drift」等 competing mechanisms。此時才需要 system-level specialist 與更多 active experiment logic。

26. Requirement-to-Test Traceability - 規格到測試的追溯矩陣

Agent 寫出很多 code 不等於系統完成。SAI 3.2 以 traceability matrix 將需求直接綁到測試；IMPLEMENTATION_STATUS.md 應引用這些 Requirement IDs 與 Test IDs。

Requirement	Test ID	Test type	Pass condition
ART-001	T-ART-001	contract	同一 bytes 得到同一 content hash/artifact identity；修改 1 byte 必須產生新 artifact_id/hash；若同一人類文件 lineage，lineage_revision 遞增並保留 previous_artifact_id。
EVI-002	T-EVI-002	unit	parser 缺欄位時輸出 UNKNOWN/NOT_REPORTED，不得生成 default scientific value。
EVI-003	T-EVI-003	integration	LLM-generated interpretation 無法被 evidence admission gate 當作 observed/simulated evidence。
EPI-001	T-EPI-001	e2e	root-cause episode 在驗證前保留至少兩個 active/competing hypotheses。
SIM-001	T-SIM-001	contract	run manifest 缺 solver/project/condit

Requirement	Test ID	Test type	Pass condition
			ions/validity 任一必要欄位即拒絕升級正式 evidence。
SIM-002	T-SIM-002	e2e	low-fidelity contradiction 只使 hypothesis → CHALLENGED，不得直接 → REJECTED。
EXT-001	T-EXT-001	architecture	新增 ToyDomain 時 core package 無 source modification； plugin registration 即可運作。
SEC-001	T-SEC-001	security	RESTRICTED_NDA context 嘗試送 external connector 時被 policy engine 阻擋並留下 audit event。
DOM-SP-001	T-DOM-SP-001	domain	Rs trend validator 只存在 Silicon Photonics DomainPack，移除 plugin 後 core 仍可啟動。
TST-001	T-E2E-SP-001	vertical e2e	RS anomaly mock fixture 完整跑通 ingest → diagnose → test → evidence → failure/heuristic candidate。
VER-001	T-VER-001	e2e	Planner 先檢查已存在 evidence / cheap

Requirement	Test ID	Test type	Pass condition
			actions；只有較便宜層不足時才升級 simulator，並記錄選擇理由。
SRC-001	T-SRC-001	contract	以 fake GitHub/Literature connectors 替換真 provider 時 SourceRouter/cognition 不需修改；normalized record schema 相同。
GH-001	T-GH-001	integration	public fixture repo 可 search/fetch；requested ref 解析到固定 commit SHA，file content hash 與 locator 被保存。
GH-002	T-GH-002	security	未授權 private repo request 被 fail-closed 並產生 audit event；不洩漏 query/private context。
GH-003	T-GH-003	epistemic	GitHub record 能被標為 technical/prior-art source；admission gate 不允許其自動冒充 measured/peer-reviewed evidence。
EPI-003	T-EPI-003	e2e	隔離 triggering attestation 後 replay，EpistemicState 投影改變且 history 保留。

Requirement	Test ID	Test type	Pass condition
EVI-004	T-EVI-004	integration	preprint/journal/ review fixture resolve 為同一 source work ; DEPENDENCE_UNK NOWN fixture contributes 0 to independent count until resolved; promotion requiring independence stays blocked.
LLM-001	T-LLM-001	contract	Hypothesis/ Critique/Relation 缺 model/prompt/bundl e provenance 時 admission fail 。
SEC-002	T-SEC-002	security	未授權 actor 被 ACL/egress gate 拒 絕；新 artifact 未分類 時預設 restricted 。
OPS-001	T-OPS-001	e2e	delayed mock job 可 suspend/resume；重 複 completion event 不建立第二個 run 。
COST-001	T-COST-001	integration	預算不足時 model/tool call 在產 生外部 side effect 前 被阻擋。
VER-002	T-VER-002	unit	Planner 對無 capability descriptor 的 backend 不可規 劃；有 descriptor 時 依 produces/requires

Requirement	Test ID	Test type	Pass condition
			match 。
VER-003	T-VER-003	unit	CostVector 可表達時間/人力/錢/license/不可逆；Planner 不依賴單一 normalized_cost 。
DOM-SP-002	T-DOM-SP-002	domain	同一 extract_cj_rs contract 通過 simulated impedance 與 measured impedance fixtures 。
EVI-005	T-EVI-005	contract	錯誤 condition schema version/write 被拒絕；ConditionMatch 回傳受版本控制 。
SRC-002	T-SRC-002	e2e	DIAGNOSIS/NOVELTY 使用不同 source policies ； high-stakes decision 未執行 inverted retrieval 時 BELIEF_REVISION 被拒絕；執行後 CritiqueReport.inverted_bundle_id 與 bundle divergence 可追溯 。
TST-002	T-SPEC-001	spec	所有 requirement IDs 唯一；每個 normative ID 至少一個 test ；test 不引用未知 ID 。

Requirement	Test ID	Test type	Pass condition
EPI-004	T-EPI-004	unit	authority fixtures cover stronger/weaker/equivalent/incomparable; required INCOMPARABLE returns NEED_HUMAN_REVIEW, auto-creates ReviewItem(AUTHORITY_CONFLICT), and blocks belief promotion/rejection.
EPI-005	T-EPI-005	contract/e2e	direct LLM status assignment is rejected; admitted RelationJudgments + policy produce event and replayable projection.
EVI-006	T-EVI-006	contract	permuted JSON key order yields same canonical bundle hash; changed attestation order/policy/query snapshot changes hash as defined.
VER-004	T-VER-004	unit	outcome outside declared OutcomeSpace 或被 explicit constraint/ValidationReport 排除時不可算 plausible ; 合法

Requirement	Test ID	Test type	Pass condition
			outcome version 可 重現 sufficiency result ◦
VER-005	T-VER-005	unit	same actions/state/policy produce identical ranking across repeated runs; tie is resolved by configured lexicographic fallback.
SEC-003	T-SEC-003	security	fixture secret is quarantined before immutable normal ingest; redacted derivative preserves audit linkage.
SEC-004	T-SEC-004	security	UNKNOWN/ COPYLEFT code fixture is blocked from generation context unless explicit policy fixture allows it.
EVI-007	T-EVI-007	integration	mixed embedding versions are never compared; dual- index migration preserves benchmark recall within threshold before cutover.
EVI-008	T-EVI-008	integration	retracted/erratum

Requirement	Test ID	Test type	Pass condition
			fixture records source status and blocks/flags major belief promotion per SourcePolicy.
OPS-002	T-OPS-002	unit/integration	queue depth/capacity changes human- review capability availability and planner earliest_available_at .
OPS-003	T-OPS-003	e2e	one episode trace reconstructs retrieval -> LLM -> job -> run -> artifact and associated cost entries.
TST-003	T-SPEC-002	spec	normative statement registry has unique Requirement ID; no unregistered hard MUST; deferred items include explicit rationale/test date.
SYS-001	T-SYS-001	architecture	static/conformance test rejects bypass from cognition directly to EpistemicState update or support arrays on

Requirement	Test ID	Test type	Pass condition
			Attestation/Hypothesis.
LLM-002	T-LLM-002	benchmark	fixed SiPho benchmark produces baseline distributions; BenchmarkPolicy schema stores metric/threshold/sample size/calibration artifacts/version; no hard gate before calibration.
EVI-001	T-EVI-001	contract/domain	Cj/Rs fixture missing unit 、normalization basis 、bias/frequency condition 或 extraction method 時 admission fail ；完整 fixture 可 round-trip 。
EPI-002	T-EPI-002	e2e/provenance	confirmed root cause query 必須 trace 到 Relation/Attestation or Observation -> Run -> Artifact ；只有 LLM statement 的 fixture 不得確認 root cause 。

26.1 Milestone Gate Order

Milestone	Scope	Exit gate
M0a Scientific Identity Foundation	Postgres migrations 、 Artifact/SourceWork/Claim/	artifact/claim identity + bundle hash + schema/spec

Milestone	Scope	Exit gate
	Observation/Attestation/Relation schema、canonical EvidenceBundle、condition schema version、CI + T-SPEC-001/002	conformance pass；無 Actor/LLM/Simulator 也可測。
M0b Execution/Governance Foundation	Actor/ACL、Budget/CostLedger、BeliefRevisionEvent + TransitionPolicy/AuthorityPolicy、trace/span、Job primitives、ReviewItem minimum schema	event replay + transition/authority tests + ACL/budget/trace contracts pass；無 DomainPack 也可測。
M1 Research Memory	Evidence ingestion、SourcePolicy、Claim/source-work resolution、GraphRepository(as_of/bounded)、InferenceProvenance、async Job suspend/resume wiring	local document/run ingest；source-work dedup；delayed mock job resumes episode；all scientific LLM calls persist bundle+provenance.
M2 Silicon Photonics Tool Layer	Silicon Photonics DomainPack、Capability registry、Lumerical mock/run_*、backend-agnostic Cj/Rs extractors、backend validity	simulated + measured fixtures use same extractor contract；license/resource queue mock pass
M3 Hypothesis Brain	6 core roles、selected Domain Specialists、PRIMARY/FAST/EMBEDDING minimum routing、stake-adaptive debate、Critic inverted retrieval	role I/O + Position/Critique contracts pass；Critic bundle divergence measurable；benchmark thresholds are calibrated and stored in BenchmarkPolicy before gate enforcement.

Milestone	Scope	Exit gate
M4 First Vertical	VS-SP-001 full loop : intent-aware retrieval → competing hypotheses → capability/cost planning → Job → evidence → belief event → failure/heuristic candidate	fixed benchmark set (not cherry-picked) demonstrates root-cause correctness and avoids unnecessary high-cost action in predefined cases
DEFERRED M5 External Evidence Expansion	ExternalSourceAdapter registry 、PaperQA-style literature adapter 、GitHubConnector (public + policy-gated private) 、progressive enrichment 、novelty audit boundary 、ref-pinned snapshot/cache ◦	external evidence enters with source/condition/rights/provenance metadata ; GitHub source 可追到 repo + resolved commit/ref ; 移除 GitHub provider 不影響 core cognition ◦
DEFERRED M6 Multi-physics Ring	MODE/INTERCONNECT + ring mechanism benchmark	end-to-end mechanism discrimination benchmark
DEFERRED M7 Operational Lab Brain	approved transcript miner, ELN/Git watchers, approvals, security modes	real lab workflow pilot with low human write friction
DEFERRED M8 Learning / Second Domain	case-based retrieval, trajectory export, second DomainPack	EXT-001 demonstrated on adjacent domain

結論 - SAI 3.2 的正確用途與 Executable Epistemic Architecture 北極星

North Star / Laboratory Innovation Brain 不應儲存「AI 說過什麼」，而應儲存「實驗室知道什麼、為什麼知道、在什麼條件下知道、哪些地方仍不確定，以及下一個哪種最低成本驗證最有價值」。

SAI 3.2 不是系統名稱，而是能交給 Agent 實作、也能讓人類理解與約束 Agent 的 System Analysis with AI 規格。它的成功標準不是「文件寫得完整」，而是能讓不同 Agent/工程師在不同時間依同一組 scientific contracts 建出相容、可驗證、可延伸的 Laboratory Innovation Brain。

如果只完成 RAG、多 Agent 與自動 Lumerical，Laboratory Innovation Brain 仍然只是工程助理；真正的核心是

Artifact/SourceWork→Claim/Observation→Attestation→RelationJudgment→TransitionPolicy→BeliefRevisionEvent→EpistemicStateProjection，配合 competing hypotheses、adversarial retrieval 與最低成本充分驗證。

Silicon Photonics 是第一個非常合適的 sandbox：物理鏈明確、多 simulator 可 objective verification、設計空間存在多目標 trade-off，又有大量歷史 paper / internal simulation / measurement 可以形成 evidence。完成第一個 ring modulator closed loop 後，Laboratory Brain core 應能在不改 memory/orchestration kernel 的前提下掛入新的 domain pack。

SAI 3.2 進一步明確化：Laboratory Innovation Brain 的成熟度不只看「能不能找答案/跑模擬」，還要看它是否知道什麼時候根本不需要跑模擬、能否安全承認未知、是否避免低精度數值假象、是否保存 backend validity，並以最低人類摩擦把 tacit knowledge 轉成可審核 operational memory。

Appendix A - Canonical Database Migrations (No Duplicate Schema)

Appendix A intentionally does NOT redefine tables.

Canonical database semantics live in §17 contracts + versioned migrations:

```
001_actors_projects.sql
002_artifacts_sourceworks.sql
003_claims_observations_attestations.sql
004_relations.sql
005_epistemic_events_transition_policies.sql
006_jobs_runs.sql
007_capabilities_costs.sql
008_conditions.sql
009_vectors.sql
010_review_observability_benchmark_policies.sql
```

Rule: Appendix A is an index only. If migration semantics conflict with §17 + Requirement IDs, CI MUST fail T-SPEC-002; Coding Agent may not choose one silently.

Appendix B - Lumerical Ring Modulator Minimum Tool Set

ID	Tool	最低輸入	最低輸出
SP-CH-01	charge_dc_sweep	project, bias range	carrier profiles, IV, convergence
SP-CH-02	charge_ac_sweep	project, DC points, f range	complex current / impedance
SP-CH-03	extract_cj_rs	run refs, geometry normalization	Cj/length, Rs/length, method
SP-CH-04	validate_access_connectivity	geometry artifact	connectivity warnings / current path
SP-MO-01	mode_solve	geometry/material/bias state	neff, loss, field artifact
SP-MO-02	eo_delta	two/more bias states	delta neff, delta loss, overlap
SP-IC-01	ring_spectrum	ring model + wavelength range	Q, FSR, resonance, ER/IL
SP-IC-02	nrz_eye	bitrate, drive states, CW wavelength	BER, Q, eye width, jitter, ER
SP-IC-03	thermal_lock	drift source + heater/control	lock error, power, transient metrics

Appendix C - Innovation Brain Evaluation Checklist

- 是否能指出 claim 的原始 artifact / locator ?
- 是否保存適用條件，而不是只存一個數值？
- 是否明確區分 observation / inference / hypothesis ?
- 是否能產生至少兩個相互可區分的 mechanisms ?
- 是否明確寫出 falsifier ?
- 下一個 verification action 是否真的能區分 predictions，而且沒有更便宜的充分證據？
- 若結果與預期不同，是否更新 hypothesis，而非修改敘事硬合理化？
- 是否記得過去相似 failure / episode ?
- 是否能在 privacy policy 下決定是否可連外？
- 若 simulator 與 measurement 不一致，是否保留 conflict 而非抹平？
- Evidence extractor 是否允許 UNKNOWN / NOT_REPORTED，而不是被 schema 逼著猜值？
- Low-authority contradiction 是否只形成 provisional challenge，並依 AuthorityPolicy / TransitionPolicy 決定 escalation ?
- 任何重大 belief revision 的 evidence 是否附可比較的 EvidenceAuthority + BackendValidity，而不是把 backend output 直接當 ground truth？
- Heuristic 是否能追到授權來源，且由人類核准後才成為 active lab rule？
- 多跳關係查詢是否經 GraphRepository，而不是散落 backend-specific query？

Appendix D - Glossary

名詞	定義
Artifact	原始或衍生的不可變研究資產；由 content hash / provenance 綁定。
Evidence	可支持/反駁 scientific proposition 的正規化證據，帶 locator 與 conditions。
Epistemic State	系統目前的科研信念狀態：hypothesis、conflict、heuristic、open question。
Research Episode	一次有 goal、hypothesis、run、observation、decision 的完整研究演繹。
Hypothesis Certificate	讓 idea 進入昂貴驗證前必須具備的 mechanism/prediction/falsifier 等欄位。
Falsifier	若觀察到某結果，應使 hypothesis 被削弱/推翻的明確條件。
Condition-aware Retrieval	先以 scientific operating conditions 判斷 comparability，再做 semantic ranking。
Typed Tool	有固定 input/output schema、provenance、validator 的工具，不讓 LLM 任意修改 evaluator。
Multi-Fidelity	依成本與需要，從 analytical/surrogate/coarse simulation 逐步升級到 high-fidelity / measurement。
Case-based Reasoning	從歷史 research episodes / failures 找相似 case 幫助當前診斷。
Novelty Auditor	與 internal reasoning 分離的外部 prior-art 檢查功能。
Claim	可被不同來源/Attestation 指向的科學命題 identity；先解 identity 再算 corroboration。
Attestation	某來源在特定 locator/conditions 下對 Claim/Observation 的具名見證。
BeliefRevisionEvent	append-only scientific state transition record；EpistemicState 由它重建。
EvidenceBundle	一次 retrieval 的可重現、可 hash 證據封裝；是 LLM inference provenance 的輸入 identity。
AuthorityPolicy	比較 EvidenceAuthority 的版本化偏序規則；可回傳 INCOMPARABLE。
TransitionPolicy	決定 hypothesis 是否允許狀態轉移的版本化規則；不是 LLM 投票。

Appendix E - Reference Repositories & Architecture Sources

來源	URL
Flexcompute AutoPhotonicDesign	https://github.com/flexcompute/autophotonicdesign
Kaimen Co-Scientist	https://github.com/Kaimen-Inc/Co-Scientist
FutureHouse PaperQA / PaperQA2	https://github.com/Future-House/paper-qa
MIT SciAgents Discovery	https://github.com/lamm-mit/SciAgentsDiscovery
Sakana AI Scientist v2	https://github.com/SakanaAI/AI-Scientist-v2
Stanford Virtual Lab	https://github.com/zou-group/virtual-lab
Agent Laboratory	https://github.com/SamuelSchmidgall/AgentLaboratory
Stanford STORM / Co-STORM	https://github.com/stanford-oval/storm
FutureHouse Robin	https://github.com/Future-House/robin
PriM	https://github.com/amair-lab/PriM
ARI - Autonomous Research Infrastructure	https://github.com/kotama7/ARI
Lumerical MCP (reference implementation)	https://github.com/leisymqaz/lumerical-mcp
Docling	https://github.com/docling-project/docling
LangGraph	https://github.com/langchain-ai/langgraph
Microsoft AutoGen	https://github.com/microsoft/autogen
pgvector	https://github.com/pgvector/pgvector
Zarr Python	https://github.com/zarr-developers/zarr-python
DuckDB	https://github.com/duckdb/duckdb
AiiDA Core	https://github.com/aiidateam/aiida-core
DVC	https://github.com/iterative/dvc
MLflow	https://github.com/mlflow/mlflow

註：本文件將這些專案視為 architecture references，不代表 Laboratory Innovation Brain 直接複製其程式碼。實際 fork / copy 前應逐一確認最新 license、版本、維護狀態與第三方依賴條款；Laboratory Innovation Brain 盡量透過 adapter / clean-room reimplementation 保持核心可替換。

Appendix F - Implementation Agent Boot Prompt

以下 Prompt 可直接給 Coding Agent 作為初始施工指令；每次只替換 CURRENT MILESTONE 與 repository state，不要刪除 architecture guardrails。

You are the implementation agent for the target system described by SAI 3.2 (System Analysis with AI).

IMPORTANT TERMINOLOGY

- SAI 3.2 is the specification document, NOT the product name.
- The target system is Laboratory Innovation Brain.
- Silicon Photonics is the first DomainPack, not a core hard-code.

BEFORE CODING

1. Read SAI 3.2, IMPLEMENTATION_STATUS.md, accepted ADRs, current milestone and Requirement/Test IDs.
2. Run spec-conformance checks. A normative conflict BLOCKS the slice; do not resolve by choosing the most code-like paragraph.
3. Implement the smallest coherent slice. M0a and M0b are separate gates.

SCIENTIFIC STATE RULES

- Canonical path: Artifact/SourceWork -> Claim/Observation -> Attestation -> RelationJudgment -> TransitionPolicy -> BeliefRevisionEvent -> EpistemicStateProjection.
- NEVER mutate EpistemicState or Hypothesis status directly.
- NEVER store evidence_for/evidence_against/support_targets arrays as parallel truth.
- Resolve source-work/claim identity before corroboration counts.
- Inference is not Evidence. Unknown is valid; do not invent missing scientific values.
- Every scientific LLM output/relation requires InferenceProvenance and canonical EvidenceBundle hash.
- Authority comparisons use AuthorityPolicy.compare(); required INCOMPARABLE MUST create ReviewItem(AUTHORITY_CONFLICT) and block promotion/rejection until reviewed.

COGNITION / MODELS

- Core cognition = 6 roles: Supervisor, Evidence Researcher, Hypothesis Engine, Adversarial Critic, Verification Planner, Novelty Auditor.
- Domain specialists are registered by DomainPack.
- Model Router exposes 6 LLM slots: PRIMARY, ADVERSARIAL, FAST, CODE, PRIVATE_LOCAL, VISION + EMBEDDING (non-LLM).
- High-stakes decisions MUST execute Critic inverted retrieval and persist inverted_bundle_id before BELIEF_REVISION. Major REJECT/irreversible actions also require an independent critique path and external evidence/verification adjudication; provider diversity alone is not truth.

VERIFICATION

- Verification Planner is domain-agnostic. Literature, analytical, historical, numerical, surrogate, simulation, measurement and human review are pluggable Capabilities.
- A candidate is sufficient only via a declared/versioned OutcomeSpace + ValidationReport-compatible plausible outcome + TransitionPolicy effect or blocking-conflict resolution.
- Use CostVector + Pareto filtering + versioned deterministic SelectionPolicy; never hide all cost in one scalar.
- Human review is a finite Capability whose availability comes from ReviewQueue capacity.
- run_* is backend-bound; extract_* is backend-agnostic.

SECURITY / OPERATIONS

- Enforce Actor/ACL, privacy/rights, secret scan, license policy and BudgetGate BEFORE side effects.
- Long-running actions are Jobs with suspend/resume, idempotency, retry/backoff and trace_id.
- External providers go through ExternalSourceAdapter/SourceRouter. Local Git provenance and remote GitHub retrieval are separate.
- Preserve repository + resolved ref/commit + content/snapshot hash for cited external code.
- Private/restricted access fails closed.
- Vector retrieval never mixes incompatible embedding spaces.

CHANGE CONTROL

- Core MUST NOT import silicon_photonics.
- No silent schema drift. Use migrations + schema versions.
- MUST change -> Requirement/Test update. Architecture boundary change -> ADR. P0 scientific/security semantics -> human approval.

- Do not add dashboard, extra DB, graph service or swarm complexity before the milestone requires it.

AFTER EACH SLICE

- Run unit/contract/integration/spec tests.
- Update IMPLEMENTATION_STATUS.md with Requirement IDs, files, tests, blockers, costs and next smallest slice.

Appendix G - Architecture Decision Record (ADR) Template

ADR-XXXX: <Decision Title>

Status: Proposed | Accepted | Rejected | Superseded

Date:

Affected Requirements:

Human Approval Required: Yes/No

Context

What problem forces a design decision?

Constraints

Scientific semantics, security, domain boundary, existing data migration, environment.

Options Considered

1. ...

2. ...

3. ...

Decision

Chosen option and why.

Consequences

Positive, negative, operational cost, scientific risks.

Migration / Rollback

How existing data/code is migrated; how to revert.

Tests / Evidence

What benchmark or test validates this decision.

Appendix H - DomainPack Interface Acceptance Checklist

Check	Required
Manifest declares domain id/version/capabilities	MUST
Condition schema can represent UNKNOWN and source status	MUST
Domain validators return reports; they do not mutate raw evidence	MUST
Domain tools register through generic ToolRegistry	MUST
Core imports no module from concrete domain package	MUST
Domain pack supplies at least one contract fixture + benchmark	MUST
Second domain can coexist without changing core entity semantics	MUST
Specialist prompts/models are replaceable and are not truth authorities	MUST
Domain-specific ontology may expand incrementally	SHOULD
Dedicated graph DB / custom model training	MAY

Appendix I - IMPLEMENTATION_STATUS.md Template

Laboratory Innovation Brain - Implementation Status

Spec: SAI 3.2

System Version: 0.x.y

Current Milestone: M0a/M0b/M1/...

Last Updated:

Environment

- Python:
- PostgreSQL:
- Lumerical availability:
- Configured LLM slots:
- Verification backends available:
- External network mode:
- Configured external source adapters:
- GitHub connector auth mode / scope:

Requirement Status

ID	Status	Files	Tests	Notes
ART-001	TODO			

Statuses: TODO / IN_PROGRESS / BLOCKED / DONE / DEFERRED

Tests

- Unit:
- Contract:
- Integration:
- E2E:
- Lumerical real-run:

Architecture Decisions

- ADR-XXXX ...

Known Risks / Technical Debt

1. ...

Next Recommended Task

<one small coherent slice, with requirement IDs>

Version Notes

版本	日期	主要決策
v1.0	2026-09-09	確立 Laboratory Innovation Brain 長期定位；Silicon Photonics 為第一 domain；採 Artifact→Evidence→Epistemic State 三層；PostgreSQL-first；typed Lumerical tools；competing hypothesis + active experiment selection；privacy modes；完整 repo tree。
v1.1	2026-09-10	Evidence Integrity & Operational Learning Revision：新增 Progressive Evidence Enrichment / Extraction Abstention、GraphRepository abstraction、Fidelity-aware Evidence Authority、Simulator Validity Metadata、Heuristic Miner + Human Approval；同步更新 schemas、repository tree、roadmap、risk register 與 acceptance criteria。
v2.0	2026-09-10	SAI = System Analysis with AI；目標系統名稱與規格名稱分離；新增 Agent-executable requirements、ADR/change control、DomainPack extension contract、Silicon Photonics vertical slice、requirement-to-test traceability 與 Agent execution templates。

版本	日期	主要決策
v2.1	2026-09-10	Verification-in-the-Loop revision : Simulation/Experiment 從 core 抽象為 Evidence-Producing Action ; 新增 Least-Cost Sufficient Verification 、 Verification Planner 、 9 cognitive role I/O contracts 、 5 LLM logical slots + 1 embedding slot 、 Evidence Source Router 、 Innovation Candidate verification/novelty dual- axis , 並更新 DomainPack / repo tree / Agent boot prompt 。
v2.2	2026-09-10	External Source Adapter / GitHub Connector revision : 補齊 dataset ingress 、 provider-agnostic connectors 、 public/private repo policy 、 commit/ref provenance 、 snapshot/cache 、 rights/security 與 connector tests 。
v3.0	2026-09-10	Executable Epistemic Architecture : Claim/Attestation 、 belief event replay 、 inference provenance 、 Actor/ACL 、 Job/Cost/Capability 、 intent-aware retrieval 、 run/extract 分離 、 6 core + N specialists 、 M0- M4 收斂 。
v3.1	2026-09-10	Specification Closure

版本	日期	主要決策
		Release：移除 v2.x/v3.0 殘留；補 AuthorityPolicy/TransitionPolicy/EvidenceBundle/Position/Critique/Plan/Review contracts、plausible outcome、deterministic SelectionPolicy、 ReviewQueue capacity、secret/license/embedding/retraction/observability requirements/tests；拆 M0a/M0b。
v3.2	2026-09-10	Release Candidate Closure：修 T-SPEC 自我一致性；補 EVI-003/SEC-001 條文與 EVI-001/EPI-002 tests；定義 INCOMPARABLE→human review、BenchmarkPolicy/OutcomeSpace/ValidationReport、 DEPENDENCE_UNKNOWN 計數、high-stakes inverted retrieval；補 migration 010、normative registry/spec lint 與懸空 FK/actor references。

v2.0 (2026-09-10)

Major SAI specification revision. Corrected terminology so SAI exclusively means System Analysis with AI document/specification, while Laboratory Innovation Brain is the target system. Added dual-audience normative language, Requirement IDs, Agent change-control and ADR rules, DomainPack extension contract, adjacent-domain strategy, Silicon Photonics VS-SP-001 vertical slice, milestone gates, requirement-to-test traceability, implementation-agent boot prompt, DomainPack acceptance checklist, and implementation status template. Repository root renamed from sai/ to laboratory-innovation-brain/.

v2.1 (2026-09-10)

Core abstraction revision. Verification-in-the-Loop replaces simulator-by-default thinking. The target system now selects among existing evidence, literature, analytical checks, historical cases, numerical analysis, surrogates, simulators, measurements, and human review using privacy/authority/cost constraints. Defined exact cognitive-role I/O and model-slot contracts for implementation agents.

v2.2 (2026-09-10)

External knowledge connectivity revision. Added ExternalSourceAdapter as the mandatory provider-agnostic boundary for literature, GitHub, patents, web, ELN/LIMS and future sources. Added GitHubConnector with public/approved-private access modes, ref/commit-pinned provenance, snapshots/cache, rights/license metadata, fail-closed security, normalized ExternalSourceRecord schema, connector requirements/tests, and explicit separation of remote GitHub retrieval from local Git provenance. Updated repository tree, risks, milestones, acceptance criteria and implementation-agent rules.

v3.0 (2026-09-10)

Executable Epistemic Architecture. Added Claim/Observation/Attestation identity and source-work independence, append-only BeliefRevisionEvent with replayable EpistemicState projection, mandatory InferenceProvenance, Actor/ACL/default-deny ingestion, Job suspend/resume, CostLedger/BudgetGate, Capability descriptors, intent-aware SourcePolicy, Critic inverted retrieval and stake-adaptive debate, versioned ConditionSchema/ConditionMatch, EvidenceAuthority abstraction, backend-agnostic metric extractors, execution trace, spec-consistency tests, and a revised M0-M4 implementation scope. Reframed cognition as 6 core roles + N DomainPack specialists.

v3.1 (2026-09-10)

Specification Closure Release. Removed stale v2.x/v3.0 normative remnants, eliminated the need for a Normative Override, and closed implementation ambiguities around evidence authority comparison, deterministic belief transitions, canonical EvidenceBundle hashing, debate object contracts, verification sufficiency/outcome-space semantics, deterministic Pareto fallback, human-review capacity, operational/security requirements, and milestone decomposition. No major architectural direction changed from v3.0.

v3.2 (2026-09-10)

Release Candidate Closure. Closed remaining self-consistency blockers in requirement/test traceability; defined Authority INCOMPARABLE review semantics, BenchmarkPolicy, OutcomeSpace, ValidationReport and deterministic disagreement contracts; made high-stakes inverted retrieval behavior mandatory; aligned repository tree, migration 010, normative registry/spec lint, FailureAnalysis/approval references and current-version labels. No major architecture changed from v3.1.